

PRIMARY 6 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	The Solar System I	Components of the solar system - Sun, planets, moons, asteroids, comets, meteors; The Sun as a star; Properties of the Sun; Importance of the Sun to life on Earth; Solar energy
2	The Solar System II	The eight planets - Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune; Order from the Sun; Characteristics of each planet; Inner planets (terrestrial) vs outer planets (gas giants); Dwarf planets (Pluto); Exploration of space
3	The Earth and Its Movements I	Structure of the Earth - crust, mantle, outer core, inner core; Shape of the Earth (sphere/geoid); Rotation of the Earth; Day and night; Time zones; Rotation period (24 hours)
4	The Earth and Its Movements II	Revolution of the Earth around the Sun; Orbit and revolution period (365¼ days); Seasons - spring, summer, autumn, winter; Equinoxes and solstices; The Earth's axis tilt (23.5°); Why Nigeria has two seasons
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Changes in Our Climate I	Weather vs climate; Elements of weather and climate - temperature, rainfall, humidity, wind, air pressure, sunshine; Climate zones - tropical, temperate, polar; Nigeria's tropical climate; Climate change - meaning and causes
8	Changes in Our Climate II	Effects of climate change - global warming, melting ice caps, sea level rise, extreme weather, drought,

		flooding; Greenhouse gases; Deforestation and climate; Adaptation and mitigation strategies; Individual and collective action
9	The Human Body System: Blood Circulation I	Components of the circulatory system - heart, blood vessels (arteries, veins, capillaries), blood; Structure of the heart - chambers, valves; How the heart works as a pump; Heartbeat and pulse; Blood pressure
10	The Human Body System: Blood Circulation II	Composition of blood - red blood cells, white blood cells, platelets, plasma; Functions of blood - transport, protection, regulation; Blood types and groups; Blood donation; Keeping the circulatory system healthy - exercise, diet, avoiding smoking
11	REVISION	Review of all topics covered; Diagrams practice; Question and answer sessions; Practical demonstrations
12	EXAMINATION	End of term assessment - theory and practical
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 6 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

FIRST TERM

WEEK 1: THE SOLAR SYSTEM I

OBJECTIVES By the end of the lesson, pupils should be able to:

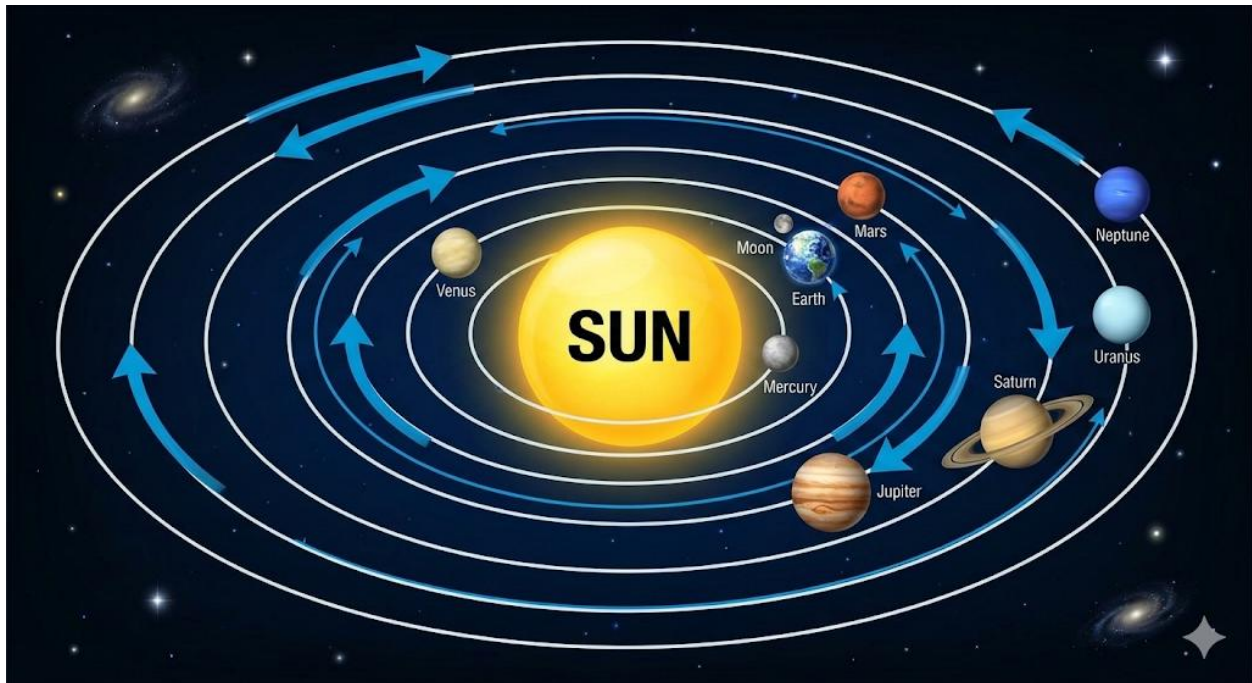
1. Define the Solar System.
2. List the main components of the solar system (Sun, planets, moons).
3. Describe the Sun as a star and its properties.
4. State three reasons why the Sun is important to life on Earth.

CONTENT

1. Meaning of the Solar System The Solar System is like a giant family in space with the Sun as the mother. It is made up of the Sun and everything that moves around it. Imagine the Sun sitting in the middle of a huge round playground. All the other objects are moving around it on different paths, like children running around a parent. These objects include:

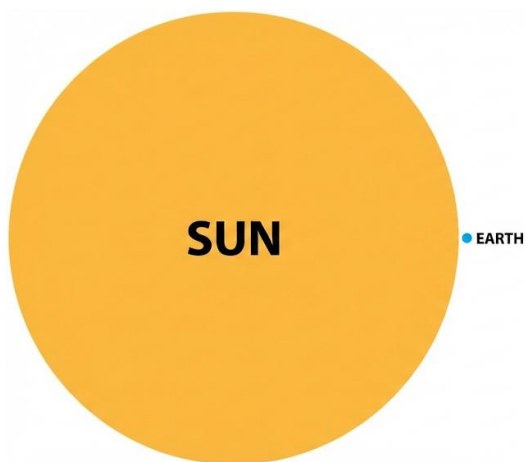
- The eight major planets.
- Smaller objects called dwarf planets (like Pluto).
- Moons, which are like smaller balls that go around the planets.
- Rocky pieces called asteroids.
- Icy balls with long tails called comets.
- Tiny rocks called meteors.

2. The Sun The Sun is a special ball of very hot, glowing gases. It is not a planet; it is a star. It looks so big and bright to us because it is the closest star to Earth. All the other stars we see at night are suns for other planets, but they are very, very far away.



What it is made of: The Sun is mostly made of two gases: hydrogen and helium. These gases are so hot that they glow and give off light and heat, like a super powerful electric bulb or a giant fire.

- **Gravity:** The Sun has a powerful invisible pull called gravity. This pull acts like an invisible rope that holds all the planets in their paths so they don't float away into darkness.
- **Size:** The Sun is enormous. If the Sun were a hollow ball, you could fit more than one million planets the size of Earth inside it! The Earth is like a tiny pepper compared to a giant watermelon (the Sun).



3. Importance of the Sun

We cannot live without the Sun. It gives us many things:

- **Heat and Warmth:** The Sun's heat keeps the Earth from being a frozen, icy ball. It makes our mornings warm.
- **Light:** Sunlight brightens our day, helping us see, play, and work. It chases away the darkness of night.
- **Food for Plants (Photosynthesis):** Plants use sunlight to make their own food. This food becomes food for animals and people. No Sun means no food.
- **Strong Bones:** When we play in the morning sunlight, it helps our skin make Vitamin D, which makes our bones and teeth strong and healthy.
- **Water Cycle:** The Sun's heat causes water from rivers and oceans to rise into the sky to form clouds, which later bring us rain.

EVALUATION

1. What is the Solar System?
2. Why is the Sun called a star and not a planet?
3. What is the name of the force that holds the planets near the Sun?
4. Give two things we get from the Sun that help plants to grow.
5. If there was no Sun, why would it be impossible for us to see during the day?

ASSIGNMENT

1. **(Drawing)** Draw a big, bright Sun and color it yellow and orange.
2. **(Oral/Written)** Tell your parent or guardian one reason why the Sun is important and write it down.
3. **(Written)** List three things in the solar system besides the Sun and the Earth (e.g., Mars, Moon, Comet).

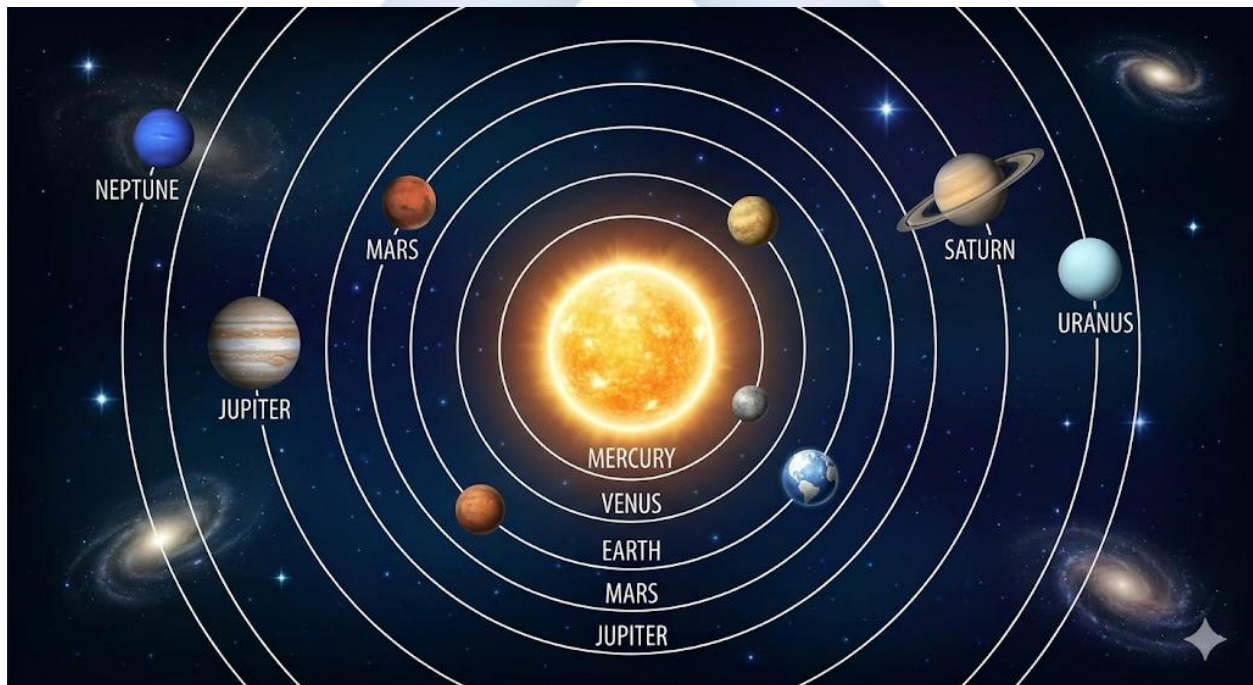
WEEK 2: THE SOLAR SYSTEM II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the eight planets in their correct order from the Sun.
2. Distinguish between Inner Planets and Outer Planets.
3. Identify the largest and smallest planets.
4. Mention the position of Pluto (Dwarf Planet).

CONTENT

1. The Eight Planets The eight planets move around the Sun on paths called orbits. To remember their order from the Sun, we use this funny sentence: "My Very Educated Mother Just Served Us Noodles." The first letter of each word matches a planet's name.



Order from Sun Planet Name Nickname / Key Fact 1st Mercury Closest to Sun. Very small and hot. 2nd Venus Hottest planet. Very bright. Earth's "twin". 3rd Earth Our home. Has water and air for life. 4th Mars The "Red Planet" due to rusty soil. 5th Jupiter Largest planet. Has a big red storm spot. 6th Saturn Famous for its beautiful rings. 7th Uranus Spins on its side. An ice giant. 8th Neptune Farthest from Sun. Very cold and windy.

2. Inner vs. Outer Planets Planets are grouped by their position and what they are made of.

- **Inner Planets (Rocky/Terrestrial):** Mercury, Venus, Earth, Mars.
 - They are closer to the Sun.
 - They are smaller and have a solid, rocky surface like the ground we walk on.
- **Outer Planets (Gas Giants):** Jupiter, Saturn, Uranus, Neptune.
 - They are very far from the Sun.
 - They are very large.
 - They are not solid. They are giant balls of gas (like very thick air) with maybe a small rocky center. You cannot land on them!

INNER PLANETS	OUTER PLANETS
 Rocky Surface Higher Density Few Moons No Rings Examples: Earth, Mars	 Gas Giants Lower Density Many Moons Ring Systems Examples: Jupiter, Saturn

3. Dwarf Planets Pluto is much smaller than the other planets. Scientists found many other small, round objects like Pluto far away. So, they created a new group called "Dwarf Planets". Pluto is now the most famous dwarf planet. It is found beyond Neptune.

EVALUATION

1. Use the memory sentence to list the eight planets in order.
2. Which planet is the largest, and which is the smallest?
3. Why are Mercury, Venus, Earth, and Mars called the Inner Planets?
4. What are Saturn's rings made of?
5. What is the main difference between a planet like Earth and a gas giant like Jupiter?

6. What happened to Pluto? What is it called now?

ASSIGNMENT

1. **(Oral)** Say the planet memory sentence to three friends.
2. **(Drawing)** Draw the planet Saturn with its rings.
3. **(Written)** Write two differences: "One way Inner Planets are different from Outer Planets is..."



WEEK 3: THE EARTH AND ITS MOVEMENTS I

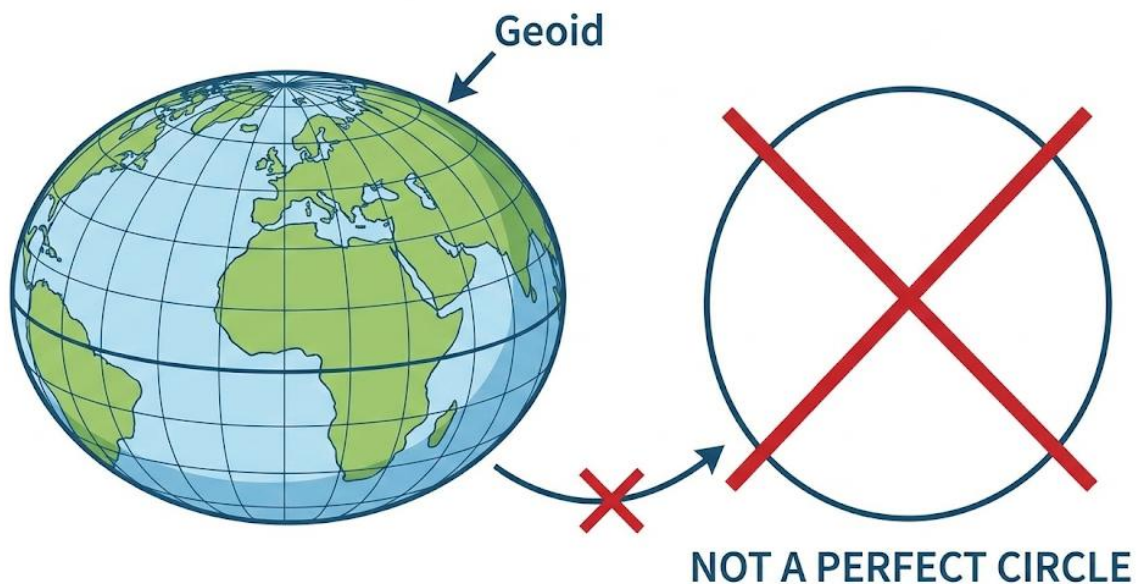
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Describe the shape of the Earth.
2. List and describe the internal layers of the Earth.
3. Define Rotation.
4. Explain the cause of Day and Night.

CONTENT

1. Shape and Structure of the Earth

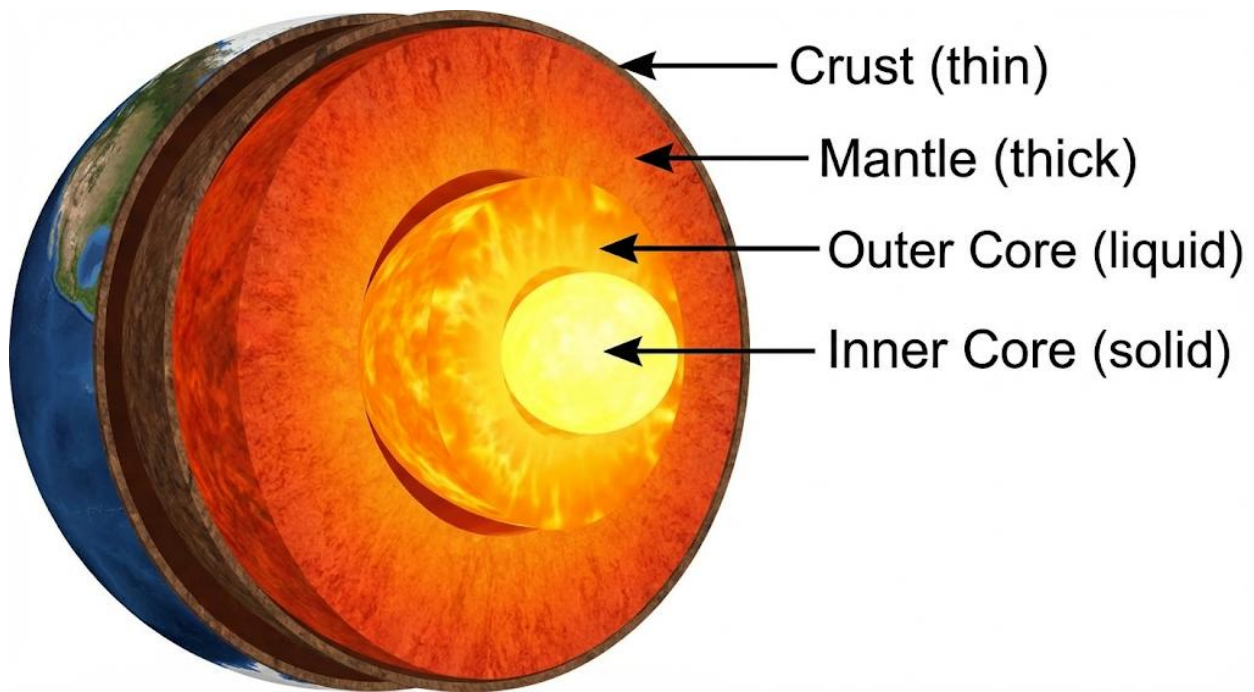
- **Shape:** The Earth is shaped like a ball, but not a perfect round ball like a marble. It is slightly squashed at the top and bottom (the poles) and bulges in the middle (the equator). This special shape is called a Geoid (say: JEE-oid). Think of a ball that is gently pressed from the top and bottom.



Internal Layers: The Earth is like an onion or a boiled egg; it has layers inside.

- **Crust:** The thin, cool, solid outside layer where we live. It's like the shell of an egg or the skin of an orange. It has land and oceans.

- **Mantle:** The thickest layer below the crust. It is made of hot, soft, solid rock that can move very slowly (like thick plastic).
- **Outer Core:** A layer of very hot liquid metal (mostly iron and nickel).
- **Inner Core:** The very center of the Earth. It is a solid ball of extremely hot metal, squeezed by all the layers above.



2. Rotation of the Earth The Earth is always spinning around like a top. This spin is called Rotation.

- It spins around an imaginary straight line called its axis. This axis goes from the North Pole (top) to the South Pole (bottom).
- One complete spin takes 24 hours, which is one full day and night.

3. Day and Night Rotation causes day and night.

- The Earth spins from west to east.
- The side of the Earth facing the Sun has daylight (morning and afternoon).
- The side facing away from the Sun has darkness (night time).
- This is why when it is daytime in Nigeria, it is nighttime in countries like America on the opposite side of the globe.

[INSERT IMAGE: A simple diagram showing a Sun on one side, lighting up half of the Earth (labeled "DAY"). The other half is in shadow (labeled "NIGHT"). An arrow shows the Earth spinning.]

EVALUATION

1. What is the name for the Earth's slightly squashed ball shape?
2. Name the four layers of the Earth from the outside in.
3. On which layer do we build our houses and farms?
4. What does "rotation" mean?
5. How long does it take for the Earth to finish one rotation?
6. If you are playing football under the Sun, is your part of the Earth facing towards or away from the Sun?

ASSIGNMENT

1. **(Drawing)** Draw and color a diagram showing the Earth's layers. Label the Crust, Mantle, and Core.
2. **(Written)** Explain in two sentences: "We have day and night because..."
3. **(Reasoning)** If it is 12 noon in Lagos, Nigeria, is it likely to be morning, afternoon, or night in London, England? (Think: Is Nigeria facing the Sun?)

WEEK 4: THE EARTH AND ITS MOVEMENTS II

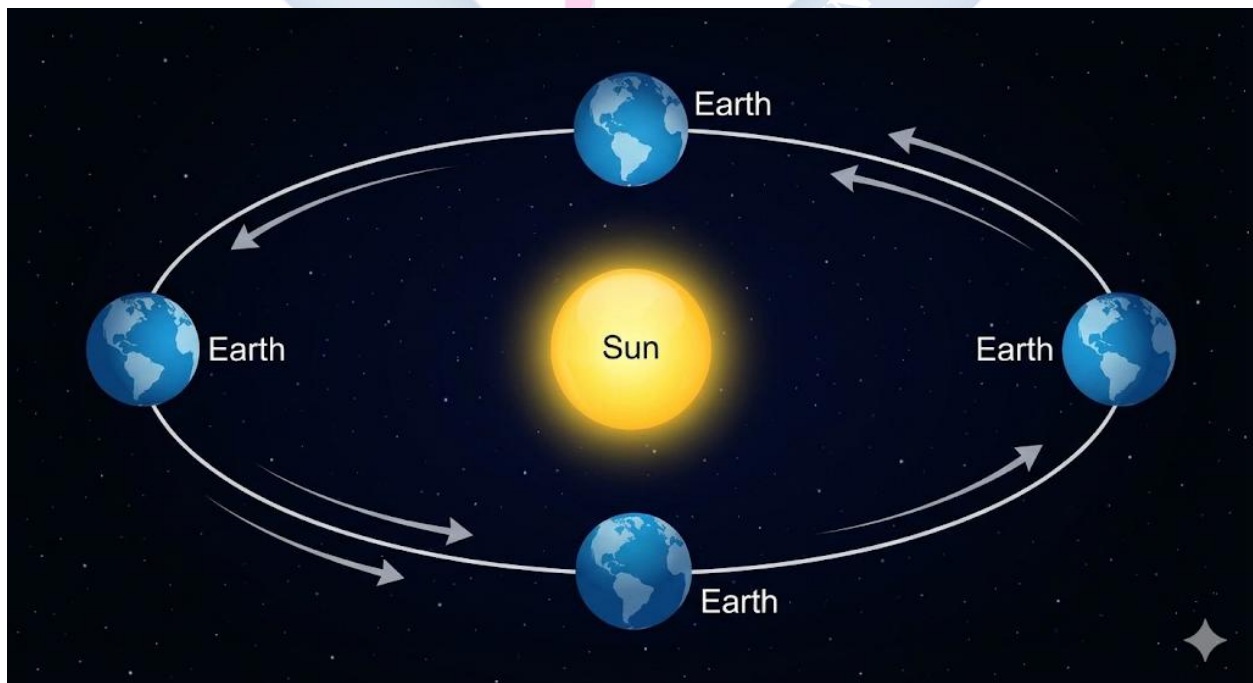
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Revolution.
2. State how long it takes Earth to revolve around the Sun.
3. Explain the cause of seasons.
4. Distinguish between the four temperate seasons and Nigeria's two seasons.

CONTENT

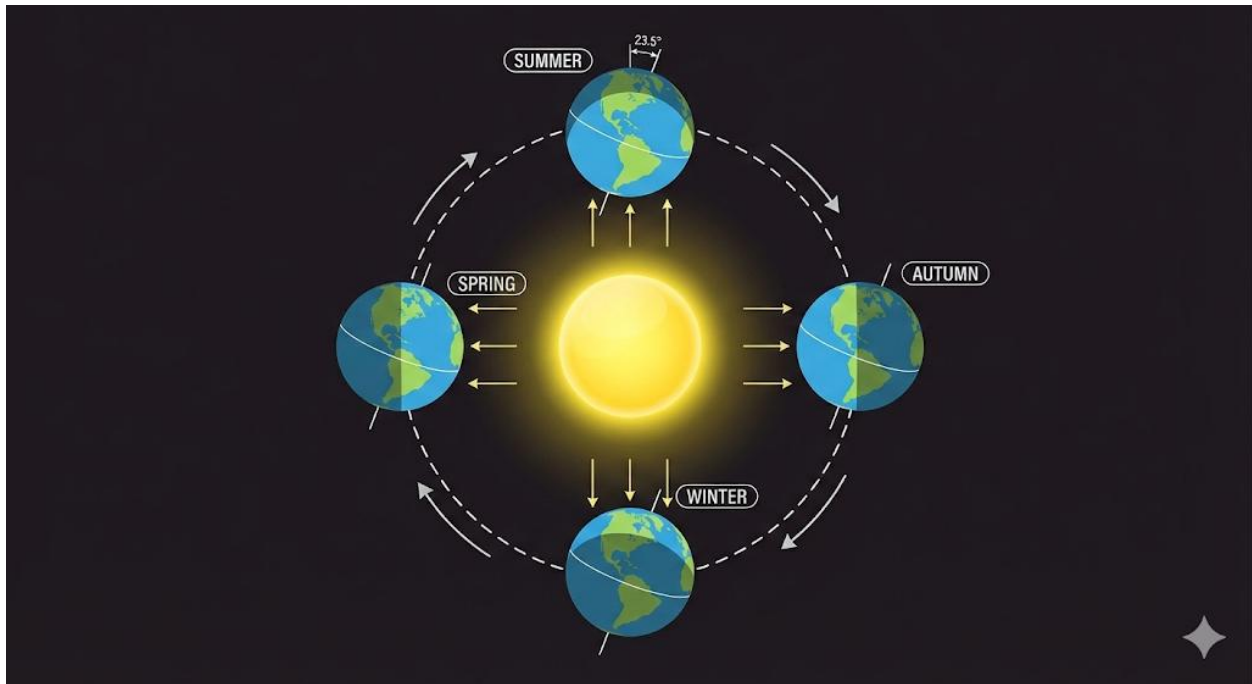
1. Revolution While the Earth spins (rotates), it is also moving in a very big, slightly oval-shaped path around the Sun. This journey around the Sun is called Revolution.

- The path it follows is called its orbit.
- One complete trip around the Sun takes 365 days and about 6 hours ($365 \frac{1}{4}$ days). This is one year.
- The extra 6 hours each year add up. After 4 years, we have an extra day, which we add to February. That year is called a Leap Year (366 days).



2. What Causes Seasons? Seasons are caused by Revolution + Tilt.

- The Earth's axis is not straight up and down; it is tilted (leaning) by about 23.5 degrees. It always leans the same way.
- As the Earth revolves, different parts of it tilt towards or away from the Sun.
- The part tilted towards the Sun gets more direct sunlight and has summer (warmer, longer days).
- The part tilted away gets less direct sunlight and has winter (colder, shorter days).



3. Seasons in Different Places

- **Temperate Regions (e.g., UK, USA):** They have four distinct seasons because the tilt affects them greatly as they go around the Sun.
 - Spring (warm, flowers bloom)
 - Summer (hot, sunny)
 - Autumn/Fall (cool, leaves fall)
 - Winter (cold, sometimes snowy)
- **Nigeria (Tropical Region):** We are near the middle of the Earth (Equator). Sunlight is strong here all year. We do not have winter or spring. We have:
 - Rainy Season (Wet): Usually April to October. Cooler with lots of rain for farming.

- Dry Season (Dry): Usually November to March. Hot, dusty, and sunny. The Harmattan wind blows in December/January.

EVALUATION

1. What is the difference between Earth's rotation and revolution?
2. How many days are in one normal year? What do we call a year with 366 days?
3. What two things work together to cause seasons?
4. Name the four seasons in countries like Britain.
5. Why does it not snow in Lagos?
6. What are the two main seasons in Nigeria?

ASSIGNMENT

1. **(Drawing)** Draw the Sun and the Earth revolving around it. Show the Earth's tilted axis.
2. **(Written)** List the months for Nigeria's Rainy Season.
3. **(Practical/Oral)** Ask an elder at home: "What do you like about the Dry Season and the Rainy Season?" Write down one answer for each.

WEEK 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7: CHANGES IN OUR CLIMATE I

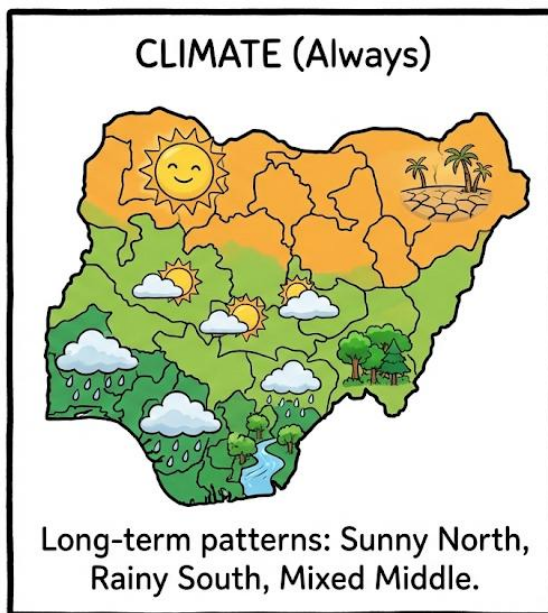
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Differentiate between Weather and Climate.
2. List the elements of weather.
3. Describe Nigeria's climate type.
4. Define Climate Change.

CONTENT

1. Weather vs. Climate

- **Weather** is about now. It is the condition of the air around us at a specific place and time. It can change quickly.
 - *Example:* "The weather today in Ibadan is sunny and hot." "There was a heavy rain this afternoon."
- **Climate** is about usually. It is the average weather pattern of a place taken over a long, long time (30 years or more).
 - *Example:* "The climate of Nigeria is generally hot and wet." "Sokoto has a hot and dry climate."



2. Elements of Weather (What we measure) These are the things we check to describe the weather:

- **Temperature:** How hot or cold the air is. Measured with a thermometer (in degrees Celsius °C).
- **Rainfall:** How much rain falls. Measured with a rain gauge.
- **Wind:** How fast and from which direction the air is moving. Measured with a wind vane (direction) and anemometer (speed).
- **Humidity:** How much water vapor (invisible gas) is in the air. High humidity feels sticky and muggy.
- **Sunshine/Sunlight:** How bright and strong the sun is.

3. Nigeria's Climate Nigeria is located in the Tropical Climate Zone, near the Equator. This means our climate is mostly hot and wet (humid) all year round. We do not experience very cold seasons like in Europe.

4. Meaning of Climate Change Climate change is a big, long-term shift in the usual weather patterns of the Earth. It means the climate is changing from what we know.

- The Earth is getting warmer on average (global warming).
- Weather is becoming more unpredictable and sometimes extreme.
 - *Example:* Heavier rains causing worse floods, or longer dry periods causing droughts.
- These changes are happening faster now mainly because of human activities.

EVALUATION

1. If I say "It is very windy today," am I talking about weather or climate?
2. Name three instruments used to measure weather and what they measure.
3. What kind of climate does Nigeria have?
4. What is the main difference between weather and climate?
5. In simple terms, what does "climate change" mean?

ASSIGNMENT

1. **(Practical/Observation)** Be a weather reporter for one day. In the morning, afternoon, and evening, look outside and write: "Time: ___, The weather is: (Sunny/Cloudy/Rainy/Windy)."
2. **(Drawing)** Draw a picture of a rain gauge collecting rain in a bottle.
3. **(Written)** Why is knowing the weather forecast useful for a) a farmer b) your mother going to the market?



WEEK 8: CHANGES IN OUR CLIMATE II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Mention three causes of climate change.
2. List three effects of climate change on the environment.
3. Define Global Warming.
4. Suggest two ways to reduce climate change.

CONTENT

1. Global Warming Global warming is the gradual increase in the Earth's average temperature. It is the main part of climate change.

- Think of the Earth wrapped in a blanket. Certain gases in the air (like carbon dioxide from smoke) act like a thicker blanket, trapping more of the Sun's heat inside. This is called the Greenhouse Effect.

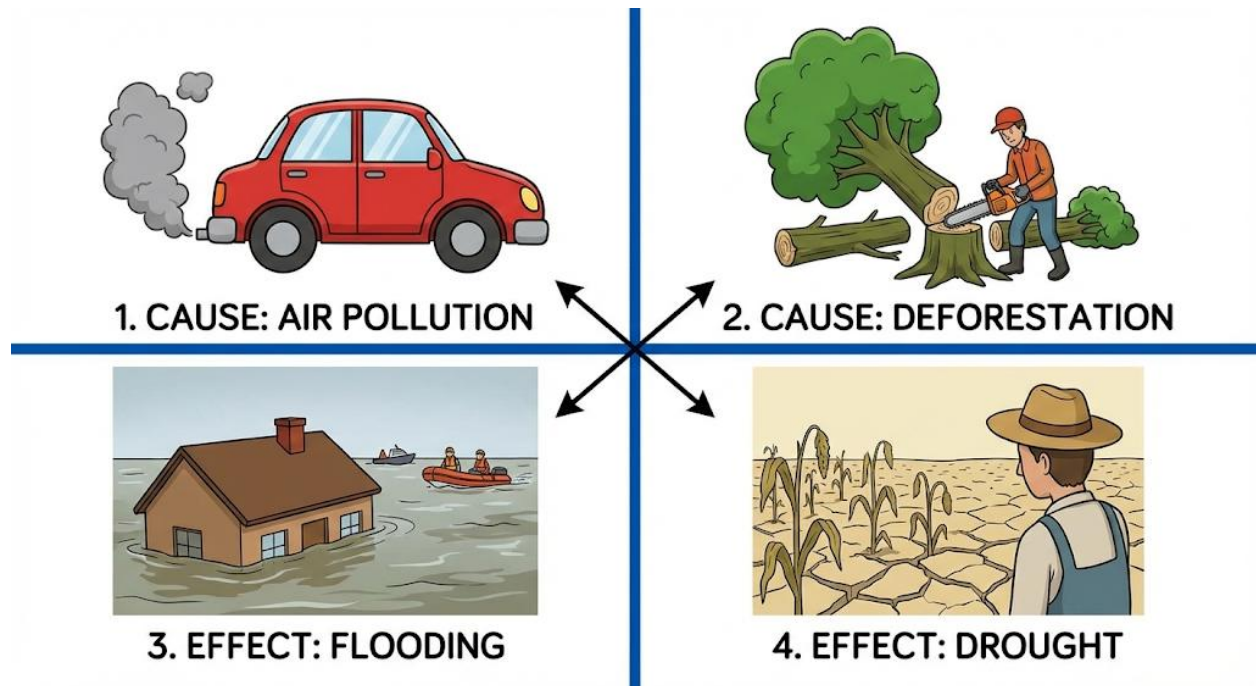
2. Causes of Climate Change (Human Activities)

- **Burning Fossil Fuels:** When we burn petrol in cars, diesel in generators, or coal in power plants, they release carbon dioxide (CO₂) gas. This is the main gas thickening the "heat blanket."
- **Deforestation:** Cutting down trees (forests) without planting new ones. Trees are like the Earth's lungs; they breathe in CO₂ and give out oxygen. Fewer trees mean more CO₂ stays in the air.
- **Bush Burning & Burning Trash:** Burning farms, forests, or plastic waste releases smoke full of harmful gases and particles.

3. Effects of Climate Change (What we see in Nigeria)

- **More Severe Flooding:** Heavy, unexpected rains can overflow rivers and destroy homes, farms, and roads.
- **Longer Droughts/Harmattan:** Dry seasons can last longer, causing water shortages and killing crops.

- **Extreme Heat:** Some days become unbearably hot, leading to heatstroke and making people sick.
- **Unpredictable Farming Seasons:** Farmers can't rely on the usual time for planting and harvesting because rains don't come as expected.



4. What Can We Do? (Solutions)

- **Plant Trees:** Trees absorb CO₂. Participate in tree-planting in your school or community.
- **Reduce, Reuse, Recycle:** Use less plastic. Reuse bottles and bags. Recycle paper and cans instead of burning them.
- **Save Energy:** Turn off lights, fans, and TVs when not in use. Walk short distances instead of using a car.
- **Spread Awareness:** Tell your family and friends about climate change.

EVALUATION

1. What is global warming?
2. Why is cutting down too many trees bad for our climate?
3. Name two human activities that cause air pollution and climate change.
4. What is one effect of climate change that can harm farmers in Nigeria?

5. How does planting trees help fight climate change?
6. What is one thing you can do at home to use less energy?

ASSIGNMENT

1. **(Drawing)** Draw a poster with the message: "PLANT A TREE, SAVE OUR FUTURE."
2. **(Practical/Written)** Look in your home. Find three items that can be recycled (e.g., plastic water bottle, old notebook, tin can). List them.
3. **(Written)** Explain why flooding can be dangerous for a community. (Think about homes, health, and food).



WEEK 9: THE HUMAN BODY SYSTEM: BLOOD CIRCULATION I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define the Circulatory System.
2. Identify the main organs of the circulatory system.
3. Describe the structure and function of the Heart.
4. Differentiate between Arteries and Veins.

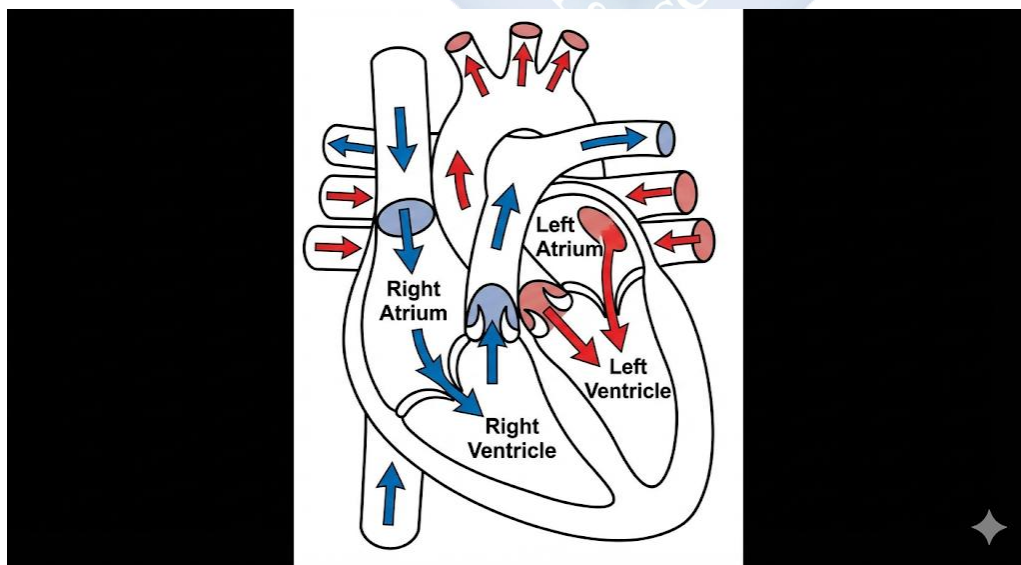
CONTENT

1. The Circulatory System This is the body's transport or delivery system. Imagine your body is a big city. The circulatory system is like the road network with trucks (blood) delivering important supplies to every house (cell).

- It carries oxygen from the lungs and nutrients from food to all parts of the body.
- It also carries away waste products (like carbon dioxide) to be removed.

2. Main Parts of the System

- **The Heart:** The powerful pump that pushes blood throughout the body.
- **The Blood:** The "delivery truck" that carries the oxygen, food, and wastes.
- **The Blood Vessels:** The "roads" or pipes (tubes) through which blood travels.



3. The Heart

- **Location:** Between your lungs, slightly to the left side of your chest. Place your hand on the center-left of your chest to feel it beat.
- **Function:** It is a muscular pump that never stops working. It contracts (squeezes) and relaxes to push blood.
- **Structure:** It has four chambers (rooms):
 - **Two Atria (Upper Chambers):** Receive blood coming into the heart.
 - **Two Ventricles (Lower Chambers):** Pump blood out of the heart to the body and lungs.
- **Pulse:** The beat you feel on your wrist or neck is the wave of blood pushed by the heart. It tells us how fast the heart is beating.

[INSERT IMAGE: A simple diagram of a heart with 4 chambers labeled: Right Atrium, Right Ventricle, Left Atrium, Left Ventricle. Show arrows for blood flow.]

4. Blood Vessels

- **Arteries:** Thick-walled, strong tubes. They carry blood **Away** from the heart. This blood is usually rich in oxygen, so we color it red in diagrams. *Example:* The artery carrying blood to your brain.
- **Veins:** Thinner-walled tubes. They carry blood **Back** to the heart. This blood has less oxygen, so we color it blue in diagrams. They have valves to stop blood from flowing backward. *Example:* Veins in your arms.
- **Capillaries:** Tiny, hair-thin vessels that connect arteries to veins. They reach every cell to drop off oxygen/food and pick up waste.

EVALUATION

1. What is the main job of the circulatory system?
2. Name the three main parts of this system.
3. What does the heart do?
4. Which type of blood vessel carries oxygen-rich blood away from the heart?

5. How many chambers does the human heart have?
6. Where on your body can you feel your pulse?

ASSIGNMENT

1. **(Practical)** Find your pulse on your wrist. Count how many beats in 30 seconds. Multiply by 2 to get your beats per minute. Write it down: "My pulse is ____ beats per minute."
2. **(Drawing)** Draw the shape of a heart and label the four chambers.
3. **(Written)** Complete: "Arteries carry blood _____ from the heart, while veins carry blood _____ to the heart."



WEEK 10: THE HUMAN BODY SYSTEM: BLOOD CIRCULATION II

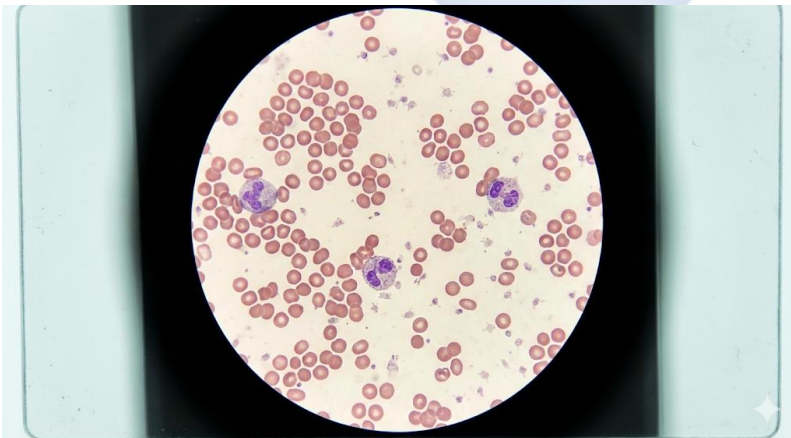
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the four components of blood.
2. State the function of Red and White blood cells.
3. Mention the four main blood groups.
4. Describe three ways to keep the heart and blood healthy.

CONTENT

1. Composition of Blood (What's inside?) Blood is not just a red liquid; it is a mixture of living cells floating in a liquid.

- **Plasma (55%):** The yellowish liquid part. It is mostly water and carries nutrients (sugar, vitamins), hormones, and waste products.
- **Red Blood Cells (Erythrocytes):** Look like tiny discs with a dent in the middle. Their job is to carry oxygen from the lungs to the body. They contain hemoglobin, a red pigment that gives blood its color.
- **White Blood Cells (Leukocytes):** The body's soldiers or defense system. They fight germs (bacteria, viruses) that cause infections and diseases.
- **Platelets (Thrombocytes):** Tiny cell fragments. They are the repair crew. When you get a cut, they rush to the spot, stick together, and form a plug (clot) to stop the bleeding.



A magnified view of a drop of blood showing many small red donut-shaped cells (RBCs), a few larger white cells (WBCs), and tiny platelets.

2. Blood Groups Not everyone's blood is the same. There are four main types or groups: A, B, AB, and O.

- It is very important to know your blood group.
- If someone loses a lot of blood (in an accident or surgery), they may need a blood transfusion (getting blood from another person). The donor's blood group must match or be compatible with the patient's, or it can be very dangerous.

3. Keeping the Heart and Circulatory System Healthy

- **Eat a Healthy Diet:** Eat plenty of fruits (oranges, watermelon), vegetables (ugu, spinach), and whole grains. Avoid too much oily/fried food, salt, and sugary drinks.
- **Exercise Regularly:** Activities like running, jumping, football, and dancing make the heart muscle strong and help blood flow easily.
- **Get Enough Rest and Sleep:** The body repairs itself during sleep. Aim for 9-10 hours of sleep as a child.
- **Avoid Harmful Substances:** Do not smoke or stay near people who are smoking. Smoking damages blood vessels and the heart.
- **Drink Plenty of Water:** Water is a major part of blood (plasma). It helps the blood flow smoothly.

EVALUATION

1. Which part of the blood fights germs and infections?
2. What gives blood its red color?
3. What is the job of platelets?
4. Name the four main blood groups.
5. Why is exercise good for your heart?
6. What is one unhealthy habit that can damage your heart and blood vessels?

ASSIGNMENT

1. **(Oral/Written)** Ask your parents or guardian: "Do you know my blood group or your blood group?" Write down the answer or write "Not Known."

2. **(Written)** List three types of food that are good for a healthy heart.
3. **(Reasoning/Written)** When you get a small cut, it bleeds and then stops. Explain what part of the blood helps to stop the bleeding and how.



PRIMARY 6 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	The Human Body System: Reproduction I	Human reproductive system - male and female; Male organs - testes, penis, sperm; Female organs - ovaries, fallopian tubes, uterus, vagina; Eggs (ova); Puberty changes in boys and girls; Age-appropriate sex education
2	The Human Body System: Reproduction II	Fertilization process; Pregnancy and development of the baby; Birth process; Growth from baby to adult; Life cycle of humans; Responsible behavior; Respecting your body; Family life education
3	Air I	Composition of air - nitrogen (78%), oxygen (21%), carbon dioxide, noble gases, water vapor; Properties of air - has mass, occupies space, exerts pressure; Importance of air to living things; The atmosphere and its layers
4	Air II	Air pollution - causes and sources; Types of air pollutants - smoke, dust, gases; Effects of air pollution on health and environment; Acid rain; Preventing and reducing air pollution; Clean air initiatives; Air quality monitoring
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Force I	Meaning of force; Types of forces - push, pull, twist; Effects of force - change in motion, shape, direction; Measuring force - Newton (N); Gravity and gravitational force; Weight vs mass; Friction - useful and harmful
8	Force II	Balanced and unbalanced forces; Pressure - meaning

		and calculation; Pressure in liquids and gases; Atmospheric pressure; Applications of pressure; Magnetic force; Electrostatic force; Force in everyday life
9	Mineral Resources I	Meaning of minerals; Types of minerals in Nigeria - petroleum (crude oil), natural gas, coal, tin, limestone, iron ore, gold, lead, zinc; Locations of minerals in Nigeria; Mining processes; Uses of minerals
10	Mineral Resources II	Importance of mineral resources to Nigeria's economy; Petroleum industry - exploration, refining, products (petrol, diesel, kerosene, etc.); Environmental impact of mining; Conservation of mineral resources; Renewable vs non-renewable resources; Sustainable use
11	REVISION	Review of all topics covered; Diagrams and experiments; Question and answer sessions; Practical demonstrations
12	EXAMINATION	End of term assessment - theory and practical
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 6 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

SECOND TERM

WEEK 1: THE HUMAN BODY SYSTEM: REPRODUCTION I

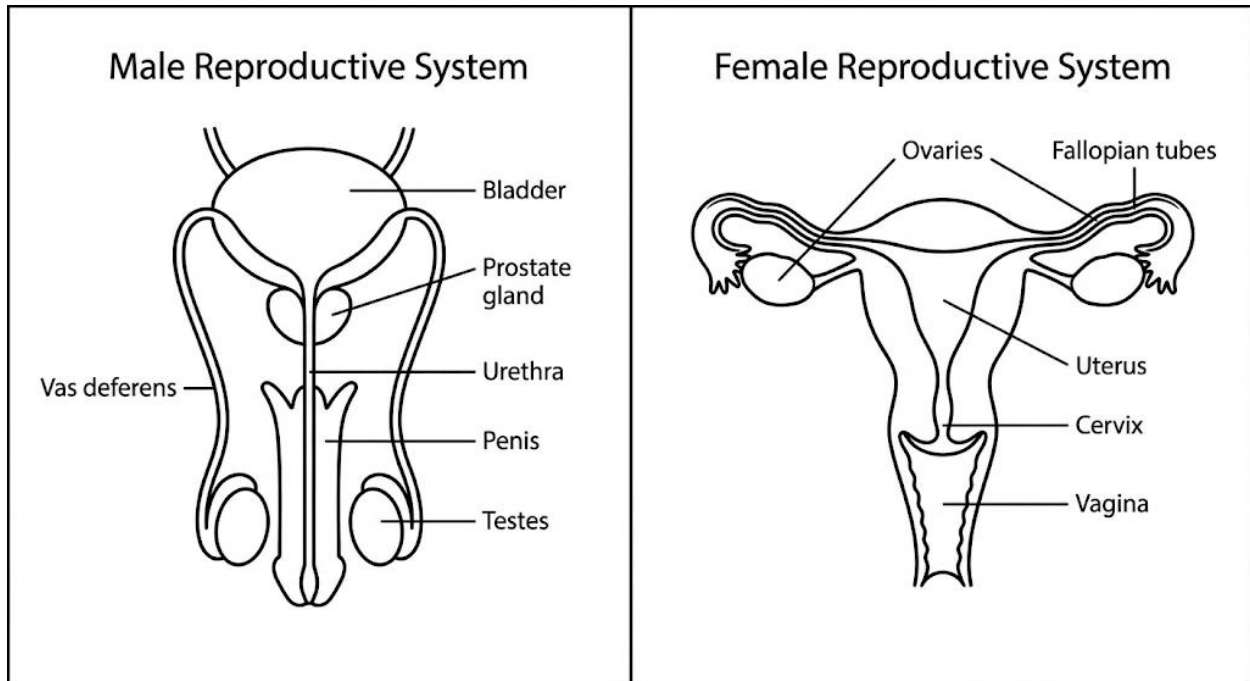
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify the main organs of the male and female reproductive systems.
2. Define puberty and list three physical changes that occur in boys during puberty.
3. List three physical changes that occur in girls during puberty.
4. Explain the importance of personal hygiene during puberty.

CONTENT

1. The Human Reproductive System All living things reproduce (have young ones). Humans have special organs in their bodies for reproduction. These organs are different for males and females.

- **Male Reproductive Organs:**
 - **Testes (Testicles):** Two oval-shaped organs inside a sac called the scrotum. They produce sperm (male reproductive cells).
 - **Penis:** The external organ through which urine and sperm leave the body.
- **Female Reproductive Organs:**
 - **Ovaries:** Two small organs that produce eggs (ova) (female reproductive cells).
 - **Uterus (Womb):** A strong, muscular bag where a baby grows and develops.
 - **Fallopian Tubes:** Two tubes that carry the egg from the ovary to the uterus.
 - **Vagina:** The birth canal; a passage leading from the uterus to the outside of the body.



2. Puberty Puberty is the time when a child's body slowly changes into an adult's body. It usually happens between ages 9 and 16. These changes are normal, natural, and happen to everyone.

3. Changes During Puberty

Changes in Boys	Changes in Girls
Voice becomes deeper (voice "breaks").	Breasts begin to develop and grow.
Hair grows on face (beard/moustache), underarms, and pubic area.	Hair grows underarms and in pubic area.
Shoulders broaden, and muscles grow bigger.	Hips become wider and more rounded.
Penis and testes grow larger.	Menstruation (monthly period) starts.

4. Importance of Personal Hygiene and Respect

- **Hygiene:** During puberty, sweat glands become more active, and body odor may develop.
 - Bathe regularly with soap and water.
 - Use deodorant or antiperspirant.

- For girls having periods: Learn to use sanitary pads or towels correctly and change them regularly to stay clean and healthy.
- **Respect:**
 - These changes are private. It is not polite to make fun of others for their changing bodies.
 - Your body belongs to you. No one should touch your private parts in a way that makes you feel uncomfortable or scared. Tell a trusted adult immediately if this happens.

EVALUATION

1. Name one organ in the male reproductive system and one in the female system.
2. Where does a baby grow inside the mother?
3. What is puberty?
4. Mention two changes that happen to a boy's voice and body during puberty.
5. Mention two changes that happen to a girl's body during puberty.
6. Why is it important to bath every day when you reach puberty?

ASSIGNMENT

1. **(Drawing/Written)** Draw a simple table with two columns: "Boys" and "Girls." List three puberty changes under each.
2. **(Written)** In your own words, explain what "menstruation" is.
3. **(Oral/Written)** Ask a trusted adult in your family: "At what age did you start noticing these body changes?" Write down the answer respectfully.

WEEK 2: THE HUMAN BODY SYSTEM: REPRODUCTION II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define fertilization.
2. Describe briefly how a baby develops in the womb.
3. Outline the stages of the human life cycle.
4. Explain what responsible behavior means regarding our bodies.

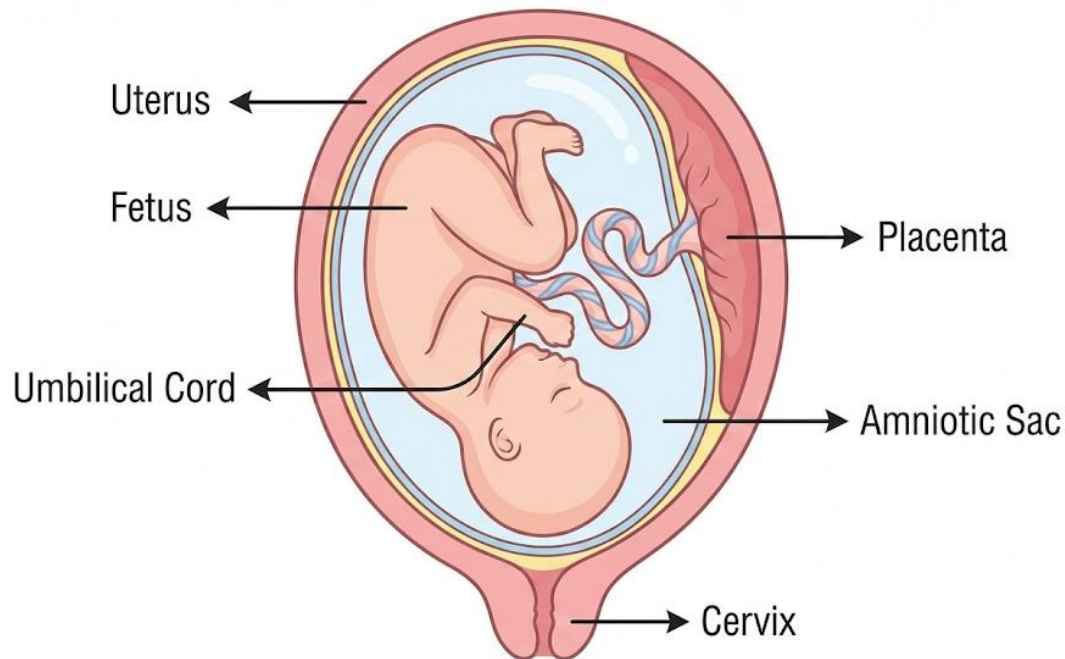
CONTENT

1. Fertilization and the Start of Life

- **Fertilization** is the joining of a sperm (from the male) and an egg (from the female). This usually happens inside the woman's fallopian tube.
- Only one sperm joins with one egg. Together they form a single new cell called a **zygote**. This is the very first cell of a new human being.

2. Pregnancy and Development

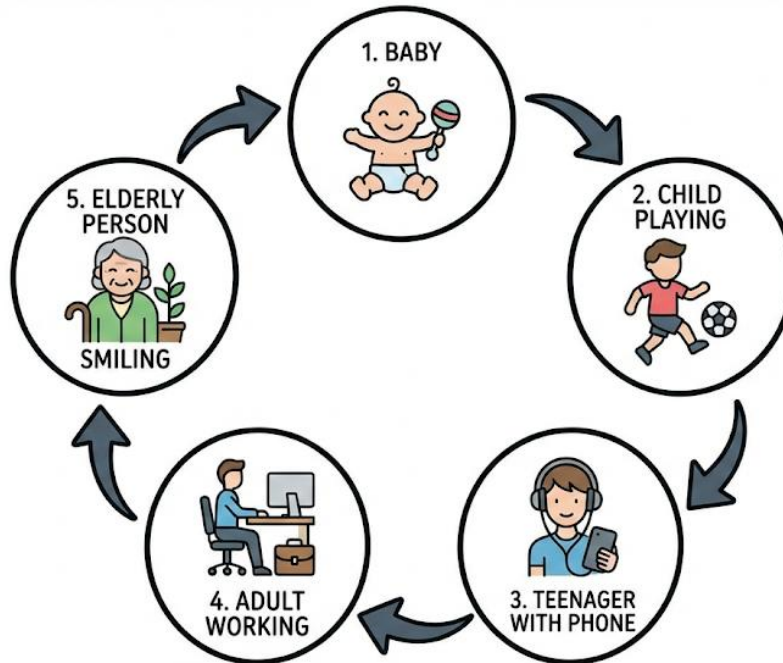
- The zygote divides into many cells, becoming an **embryo**. It travels to the uterus (womb) and attaches itself to the wall.
- After about 8 weeks, it is called a **fetus**.
- The fetus grows inside the womb for about 9 months (40 weeks).
- It is connected to the mother by the **umbilical cord**, which delivers food and oxygen from the mother's blood and takes away waste.
- The fetus is protected by a bag of fluid called the **amniotic sac**.



3. Birth When the baby is fully developed, the muscles of the uterus begin to squeeze (contractions). The baby is pushed out through the vagina. This process is called childbirth or delivery.

4. The Human Life Cycle Every person goes through these stages:

1. **Infancy (0-2 years):** Baby is totally dependent, learns to smile, crawl, walk.
2. **Childhood (3-12 years):** Grows rapidly, learns to talk, read, write, play. (You are here!).
3. **Adolescence (13-19 years):** Puberty occurs. Transition from child to adult.
4. **Adulthood (20-65 years):** Body is fully mature. People may work, marry, have children.
5. **Old Age (65+ years):** Body slows down, hair turns grey, more rest is needed.



5. Responsible Behavior

- **Respect for Self and Others:** Understand that your body is precious. Avoid risky behaviors. Treat others with respect and do not pressure them.
- **Family Planning:** Having and raising a child is a huge responsibility that requires love, maturity, stability, and resources (money, home). It is best planned for within a stable family setting when one is an adult.

EVALUATION

1. What is fertilization?
2. How long does a normal human pregnancy last?
3. What is the name of the cord that feeds the baby in the womb?
4. List the five stages of the human life cycle in order.
5. During which stage does puberty mostly occur?
6. Why is it important for people to be adults before having children?

ASSIGNMENT

1. **(Drawing)** Draw a simple timeline from "Baby" to "Old Age" and label the stages.
2. **(Written)** Write one sentence: "A baby needs its mother during pregnancy because..."

3. **(Written)** List two ways a teenager can show responsibility (e.g., focusing on studies, respecting elders).



WEEK 3: AIR I

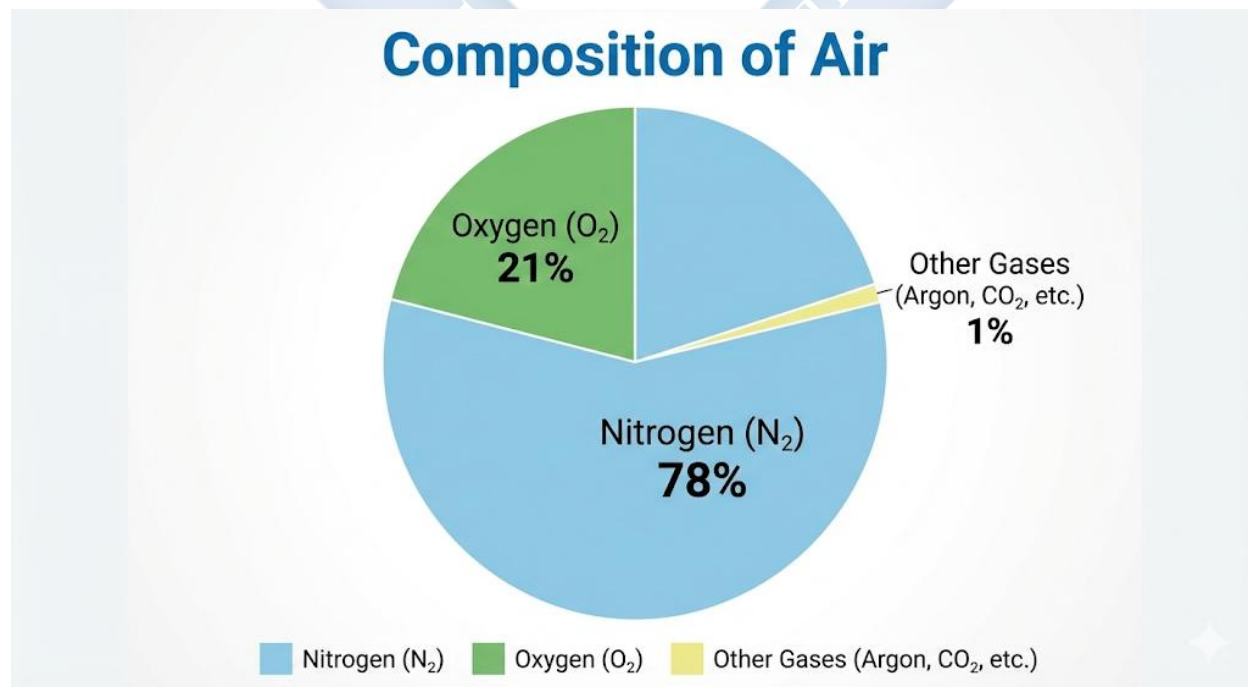
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the gases that make up the air and their percentages.
2. Mention three properties of air.
3. Explain the importance of Oxygen and Carbon Dioxide to living things.
4. Describe the atmosphere.

CONTENT

1. Composition of Air Air is a mixture of invisible gases that surrounds the Earth. We cannot see it, but we can feel it when the wind blows.

Gas Name	Percentage	Importance / Fact
Nitrogen	78% (Most!)	Dilutes oxygen; important for plants.
Oxygen	21%	Needed by humans/animals to breathe.
Others (Argon, CO ₂ , etc.)	1%	Includes Carbon Dioxide for plants.



2. Properties of Air (What air can do)

- **Air Occupies Space:** Air takes up room. Blow into a deflated balloon; it expands because you are filling it with air.
- **Air Has Weight (Mass):** A balloon filled with air is slightly heavier than an empty one. Air is not "nothing."
- **Air Exerts Pressure:** Air pushes on everything from all directions. This is why a sucked-up straw doesn't collapse and why plastic bottles crush when air is removed.
- **Air Can Be Compressed:** Air can be squeezed into a smaller space. *Example:* Pumping air into a bicycle tyre or football.

3. Importance of Key Gases

- **Oxygen (O₂):**
 - *For Breathing:* Animals and humans breathe in oxygen. Our cells use it to "burn" food and release energy.
 - *For Burning:* Fire needs oxygen to burn (combustion). Try covering a burning candle with a jar; it goes out when the oxygen is used up.
- **Carbon Dioxide (CO₂):**
 - *For Plants:* Plants take in CO₂ from the air to make their own food through photosynthesis. They give out oxygen as a waste product. We help each other!
- **Nitrogen (N₂):** It is absorbed by plants from the soil (with the help of bacteria) to make proteins for growth.

4. The Atmosphere The atmosphere is the thick layer of air surrounding the Earth, held by gravity.

- It acts like a blanket, keeping the Earth warm enough for life.
- It acts like a shield, protecting us from harmful rays from the sun and from meteors (most burn up in the atmosphere).

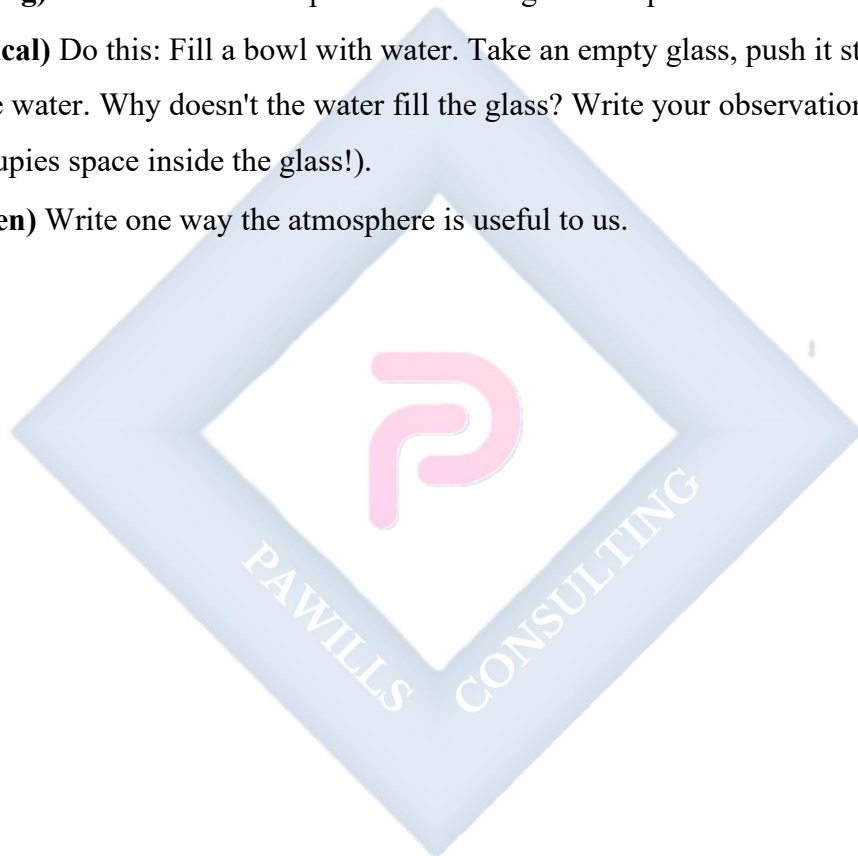
EVALUATION

1. Which gas makes up most of the air we breathe?

2. What percentage of air is oxygen?
3. Give an example that shows "air occupies space."
4. Why do animals need to breathe in oxygen?
5. What gas do plants take in to make their food?
6. What is the name of the layer of air around the Earth?

ASSIGNMENT

1. **(Drawing)** Draw and color the pie chart showing the composition of air.
2. **(Practical)** Do this: Fill a bowl with water. Take an empty glass, push it straight down into the water. Why doesn't the water fill the glass? Write your observation. (It's because air occupies space inside the glass!).
3. **(Written)** Write one way the atmosphere is useful to us.



WEEK 4: AIR II (AIR POLLUTION)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Air Pollution.
2. Identify three common sources of air pollution.
3. State three effects of air pollution on human health.
4. Suggest two ways to prevent air pollution.

CONTENT

1. Meaning of Air Pollution Air pollution happens when harmful or too many unwanted substances get into the air, making it dirty, unsafe, and unhealthy to breathe. These unwanted substances are called pollutants.

2. Sources of Air Pollutants (Where do they come from?)

- **Vehicles:** Cars, buses, motorcycles, and trucks release smoke from their exhaust pipes. This smoke contains poisonous gases and tiny particles.
- **Factories and Industries:** Big chimneys from factories release lots of smoke, dust, and chemicals into the air.
- **Generators:** Many homes and shops use petrol/diesel generators which produce thick, dark smoke.
- **Burning Activities:** Burning trash (especially plastics), burning bushes for farming, and burning firewood release harmful smoke.
- **Dust:** From unpaved roads, construction sites, and Harmattan wind.



3. Effects of Air Pollution

- **On Health:**
 - Breathing problems, coughing, and asthma.
 - Lung diseases and chest pain.
 - Eye irritation and watering.
- **On Environment:**
 - **Acid Rain:** When pollutants mix with rain water, the rain becomes acidic. This acid rain can damage plants, soil, buildings, and statues.
 - **Global Warming:** Pollutants like carbon dioxide trap heat in the atmosphere, making the Earth hotter (climate change).
 - **Haze/Smog:** A thick, dirty fog that reduces visibility and makes the air look brownish.

4. Prevention and Solutions

- **Plant More Trees:** Trees absorb carbon dioxide and give us clean oxygen. They also trap dust.
- **Use Cleaner Energy:** Use solar panels instead of noisy, smoky generators where possible.

- **Reduce, Reuse, Recycle:** Don't burn plastic waste. Reuse items or take them for recycling.
- **Maintain Vehicles:** Get your car or generator serviced regularly so it produces less smoke.
- **Use Public Transport/Walk:** For short distances, walk or cycle. This reduces the number of cars on the road.

EVALUATION

1. What is air pollution?
2. Mention two things in your community that cause air pollution.
3. How can dirty air make a person sick?
4. What is acid rain?
5. Why is planting trees a good way to fight air pollution?
6. What should you do with plastic bottles instead of burning them?

ASSIGNMENT

1. **(Drawing)** Draw a poster showing the negative effects of air pollution. You can draw a child coughing near a factory.
2. **(Written)** List three sicknesses that can be caused by breathing polluted air.
3. **(Practical/Written)** Suggest one thing your family can do this week to reduce air pollution at home (e.g., stop burning trash, walk to the nearby shop).

WEEK 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7 : FORCE I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Force.
2. List three types of forces.
3. Mention three effects of force on an object.
4. Explain the difference between Mass and Weight.

CONTENT

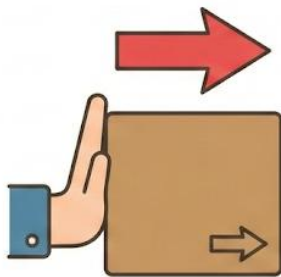
1. Meaning of Force

A force is simply a push or a pull. We use force to do almost everything. We cannot see force, but we can see what it does to objects.

- **Unit:** Force is measured in Newtons (N).

2. Types of Forces

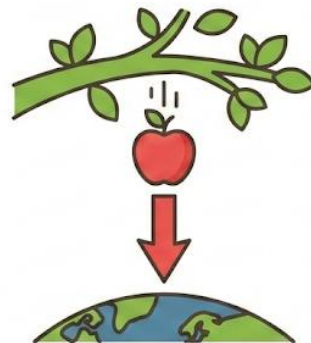
- **Contact Forces:** Forces that happen when objects touch each other.
 - *Example:* Pushing a door to open it (push). Pulling a drawer to open it (pull). Kicking a ball.
- **Friction:** A force that slows down or stops a moving object when it rubs against another surface.
 - *Example:* The brakes on a bicycle use friction to stop the wheels. Walking without slipping is possible because of friction between your shoes and the ground.
- **Gravity (Gravitational Force):** A non-contact force. It is an invisible pull that attracts all objects towards the center of the Earth. It is why anything you drop falls down.
 - *Example:* A mango falling from a tree. You staying on the ground and not floating away.



CONTACT FORCE



FRICTION



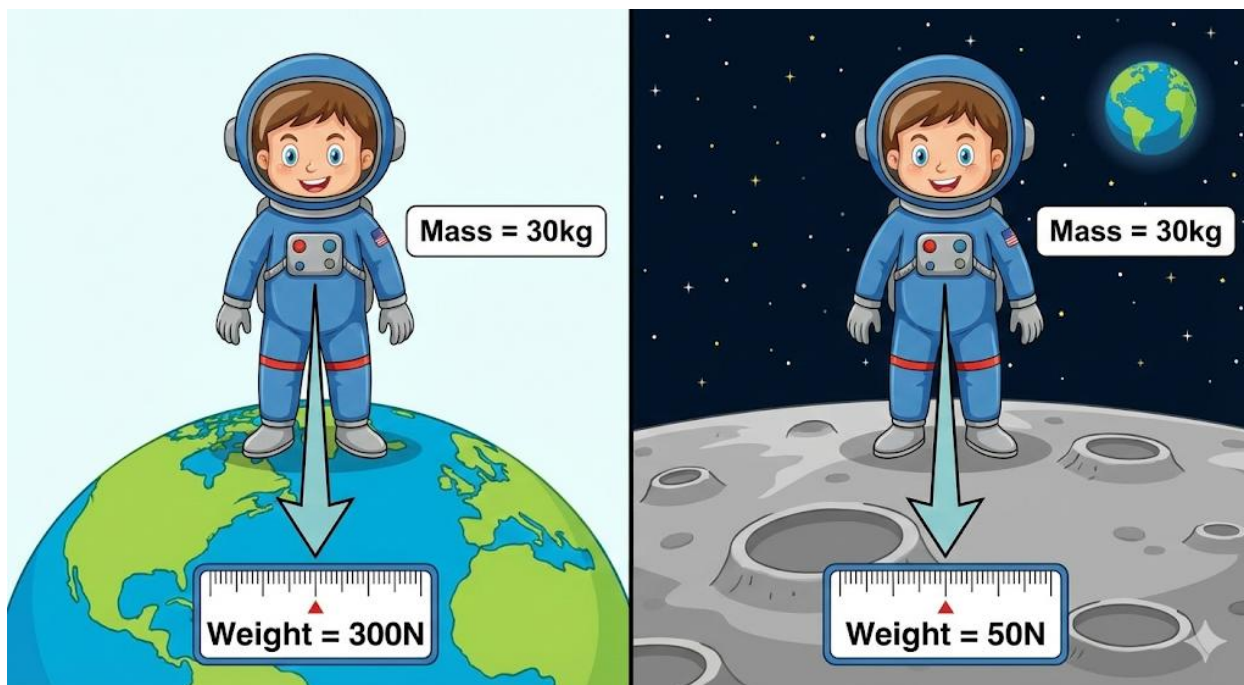
GRAVITY

3. Effects of Force Force can change an object's:

- **Movement:** It can make a stationary object start moving (e.g., pushing a stationary bicycle).
- **Speed:** It can make a moving object move faster (push it more) or slow down (apply friction/brakes).
- **Direction:** It can change the direction of a moving object (e.g., hitting a tennis ball with a racket).
- **Shape:** It can change the shape of an object (e.g., squeezing a foam, stretching a rubber band, bending a ruler).

4. Mass vs. Weight

- **Mass:** This is the amount of matter (stuff) inside an object. It stays the same everywhere. Measured in kilograms (kg). *Example:* Your body has a certain mass.
- **Weight:** This is the force of gravity pulling on that mass. It can change if gravity changes. Measured in Newtons (N).
 - You would weigh less on the Moon because the Moon's gravity is weaker than Earth's, but your mass would be the same.



EVALUATION

1. What is a force? Give one example of a pulling force.
2. What is the unit for measuring force?
3. Name the force that pulls everything towards the ground.
4. What effect does force have when you squeeze a sponge?
5. If you go to the Moon, will your mass change? Will your weight change?
6. What force tries to stop a sliding box on a rough floor?

ASSIGNMENT

1. **(Practical)** Try to push a heavy book on a smooth table, then on a rough carpet. Write which was easier and why.
2. **(Observation/Written)** Drop a flat sheet of paper and a crumpled paper ball from the same height. Which hits the ground first? Why? (Hint: think about air and gravity).
3. **(Written)** List three things you did this morning that involved a push or a pull.

WEEK 8: FORCE II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the concept of Pressure.
2. State the formula for Pressure.
3. Mention two everyday applications of pressure.
4. Differentiate between magnetic and electrostatic force.

CONTENT

1. Balanced and Unbalanced Forces

- **Balanced Forces:** When two equal forces act on an object in opposite directions, they cancel each other out. The object does not move.
 - *Example:* A game of tug-of-war where both teams are pulling equally hard. The rope stays in the middle.
- **Unbalanced Forces:** When one force is stronger than the other, the forces are unbalanced. The object moves in the direction of the stronger force.
 - *Example:* In tug-of-war, if one team pulls harder, the rope moves towards them.

2. Pressure Pressure is the amount of force applied over a certain area (space).

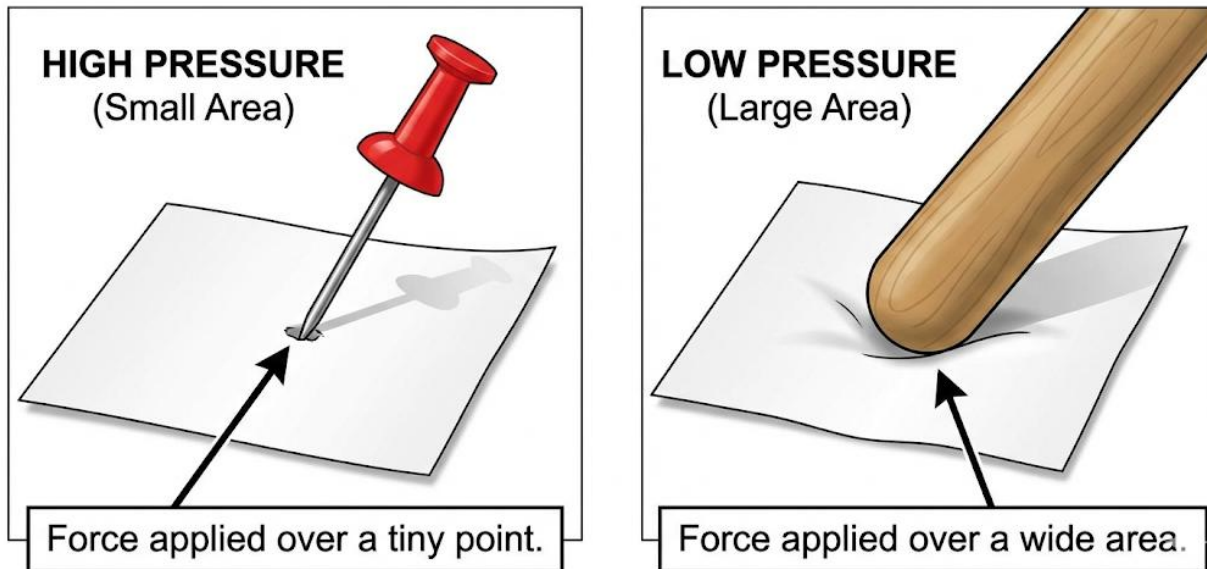
- **Formula:** $\text{Pressure} = \text{Force} \div \text{Area}$
- **Simple Rule:** The same force applied on a smaller area creates higher pressure. Applied on a larger area, it creates lower pressure.
 - *Example 1 (High Pressure):* The pointed tip of a nail has a very small area. When you hit it with a hammer (force), the high pressure allows it to pierce wood easily.
 - *Example 2 (Low Pressure):* A school bag has wide straps. They spread the weight (force) of the books over a larger area on your shoulder, reducing pressure and making it more comfortable.

3. Everyday Applications of Pressure

- **Knives and Needles:** Sharp edges have very small areas for high pressure to cut or pierce.

- **Tyre Treads:** The grooves on car tyres increase the area in contact with the road for better grip (traction) and to avoid slipping.
- **Foundation of Buildings:** Wide foundations spread the building's weight over a large area so it doesn't sink into the ground.

PRESSURE COMPARISON: Pin vs. Stick



4. Other Non-Contact Forces

- **Magnetic Force:** The force exerted by a magnet. It can attract objects made of iron, steel, nickel, or cobalt without touching them.
 - **Poles:** Every magnet has a North (N) and South (S) pole.
 - **Rule:** Like poles **REPEL** (push away). Unlike poles **ATTRACT** (pull together).
- **Electrostatic Force:** The force caused by static electricity.
 - *How it happens:* When you rub certain materials together (e.g., plastic ruler on hair, balloon on sweater), electrons move, creating a charge.
 - *Effect:* The charged object can attract small, light objects like bits of paper or make your hair stand up.

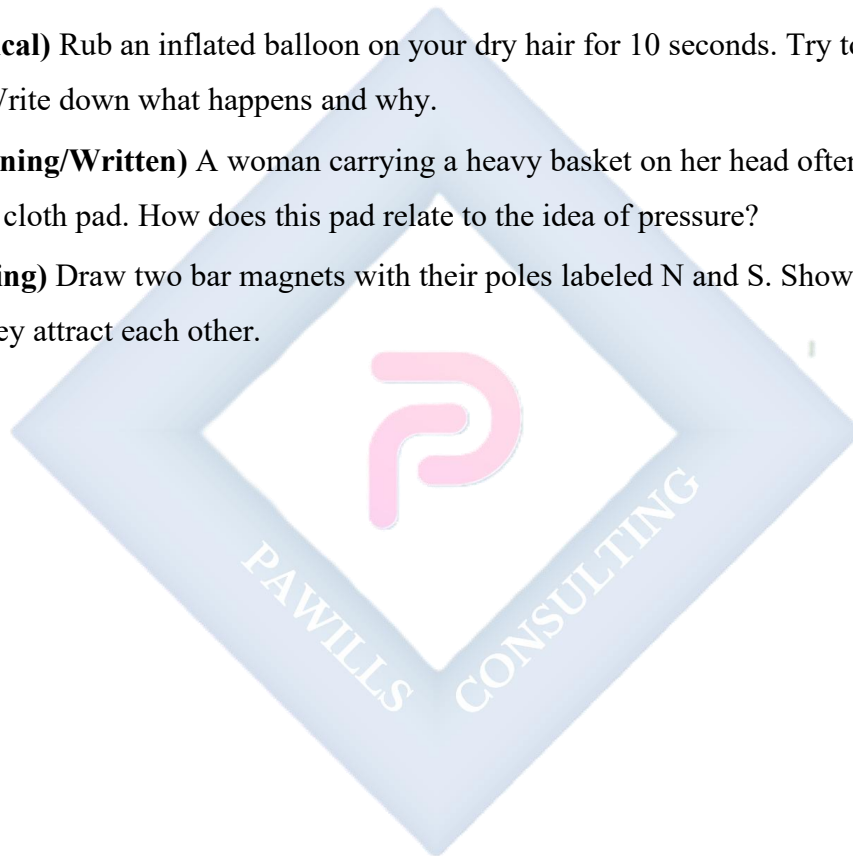
EVALUATION

1. What happens to an object when balanced forces act on it?

2. What is pressure?
3. Why do tractors have very wide tyres?
4. What will happen if you bring the North pole of one magnet near the South pole of another?
5. Give an example of something that uses high pressure to work.
6. What force makes your hair stick to a plastic comb after combing on a dry day?

ASSIGNMENT

1. **(Practical)** Rub an inflated balloon on your dry hair for 10 seconds. Try to stick it to a wall. Write down what happens and why.
2. **(Reasoning/Written)** A woman carrying a heavy basket on her head often uses a ring-shaped cloth pad. How does this pad relate to the idea of pressure?
3. **(Drawing)** Draw two bar magnets with their poles labeled N and S. Show with arrows how they attract each other.



WEEK 9: MINERAL RESOURCES I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define mineral resources.
2. List five mineral resources found in Nigeria.
3. Match specific minerals to the states where they are found.
4. Describe the mining process simply.

CONTENT

1. Meaning of Mineral Resources

Mineral resources are valuable, naturally occurring solid, liquid, or gaseous materials found in the Earth's crust. They are not made by man. They are extracted (dug out) and used to make useful products and to earn money for the country.

2. Minerals in Nigeria and Their Locations Nigeria is blessed with many mineral resources.

Mineral Resource	Where it is Found (States)	What it is Used For
Crude Oil (Petroleum)	Niger Delta (Rivers, Delta, Bayelsa, Akwa Ibom)	Fuel (Petrol, Diesel), Plastic, Chemicals
Coal	Enugu State	Fuel for power, heating
Limestone	Ogun (Ewekoro), Benue, Cross River	Making Cement
Tin & Columbite	Jos, Plateau State	Electronics, metal plating
Iron Ore	Itakpe, Kogi State	Making Steel (for cars, buildings)
Gold	Osun, Zamfara, Niger States	Jewelry, electronics, coins
Bitumen	Ondo, Lagos, Edo States	Road construction (asphalt/tar)



3. The Mining Process

Getting minerals from the ground involves steps:

1. **Exploration/Prospecting:** Geologists study the land to find where the minerals are located. They use maps, drills, and special tools.
2. **Extraction (Mining):** This is the actual digging. It can be:
 - **Surface Mining:** Digging open pits or quarries (for limestone, clay).
 - **Underground Mining:** Digging tunnels deep into the earth (for coal, tin).
3. **Processing/Refining:** The raw mineral (called ore) is cleaned, crushed, and treated to remove impurities and get the pure, useful material. Crude oil is refined in a refinery.

EVALUATION

1. What are mineral resources?
2. Which state in Nigeria is famous for Coal mining?
3. Which mineral is needed to make cement?
4. What is the most important mineral resource for Nigeria's money?
5. Name one state where Gold is found.
6. What is the name of the process of digging out minerals from the earth?

ASSIGNMENT

1. **(Drawing)** On an outline map of Nigeria, draw and label where you can find: Oil, Coal, and Limestone.
2. **(Written)** What is "Bitumen" used for? Why is it important for our roads?
3. **(Oral/Written)** Find out: What is one mineral resource that can be found in or near your own state? Ask an elder or teacher.



WEEK 10: MINERAL RESOURCES II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the importance of mineral resources to Nigeria's economy.
2. List three products obtained from refining petroleum.
3. Identify negative impacts of mining on the environment.
4. Distinguish between renewable and non-renewable resources.

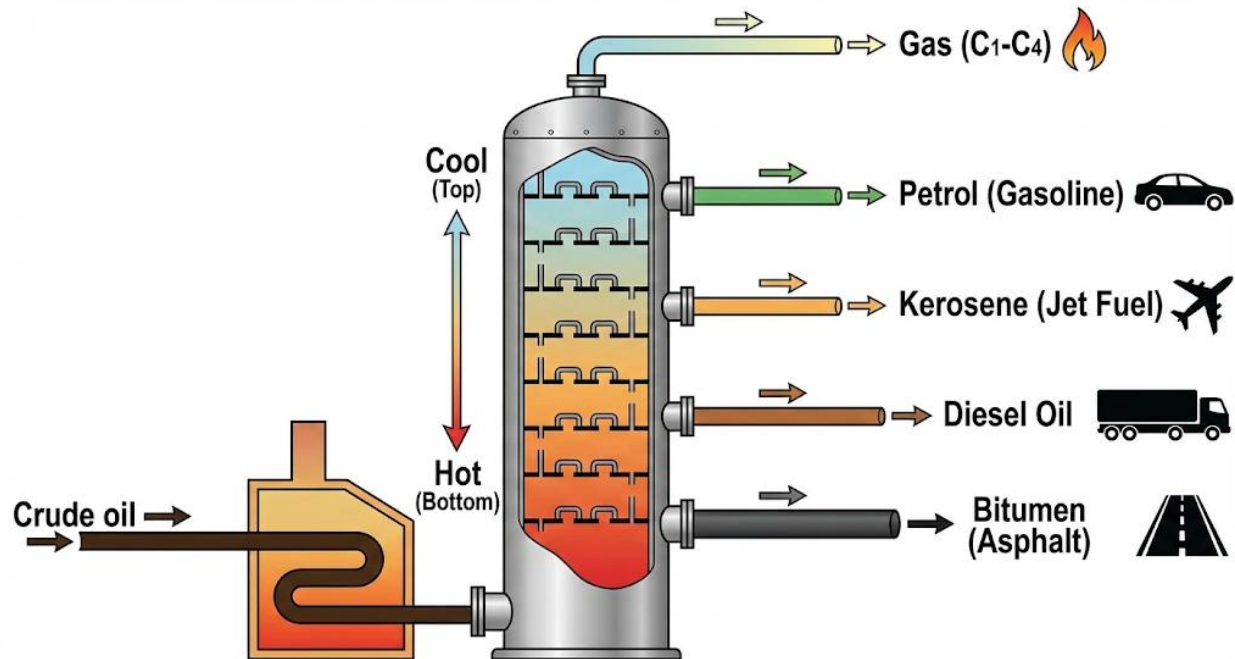
CONTENT

1. Importance of Mineral Resources to Nigeria

- **Revenue (Money):** Selling crude oil and other minerals to other countries is Nigeria's main source of foreign exchange (money). This money is used to build roads, schools, and hospitals.
- **Employment:** The mining and oil industries provide jobs for millions of people as engineers, geologists, drivers, security, and laborers.
- **Raw Materials for Industries:** Minerals supply local factories. *Example:* Limestone for cement factories, clay for brick and ceramic factories.
- **Energy Source:** Coal and natural gas are used to generate electricity in power plants.
- **Infrastructure Development:** Bitumen is used for road construction. Iron ore is used for steel to build bridges and railways.

2. The Petroleum Industry Crude oil is taken to a refinery. Here, it is heated in a tall tower called a fractionating column. Different products separate at different temperatures.

- **Petrol (Gasoline):** Fuel for cars and motorcycles.
- **Diesel (Gas Oil):** Fuel for trucks, buses, and heavy generators.
- **Kerosene:** Used for cooking stoves and lanterns in many homes.
- **Liquefied Petroleum Gas (LPG):** Cooking gas in cylinders.
- **Bitumen (Asphalt):** Used for tarring roads.



3. Negative Impacts of Mining/Oil Extraction

- **Land Degradation:** Open-pit mining leaves huge, ugly holes and destroys farmland and forests.
- **Water Pollution/Oil Spills:** Oil pipelines can leak, spilling oil into rivers and creeks. This kills fish, destroys drinking water, and harms the ecosystem of the Niger Delta.
- **Air Pollution:** Gas flaring (burning off excess gas from oil wells) releases poisonous gases and causes acid rain and respiratory illnesses.
- **Noise and Vibration:** Blasting during mining causes noise pollution and can crack nearby buildings.

4. Conservation: Renewable vs. Non-Renewable

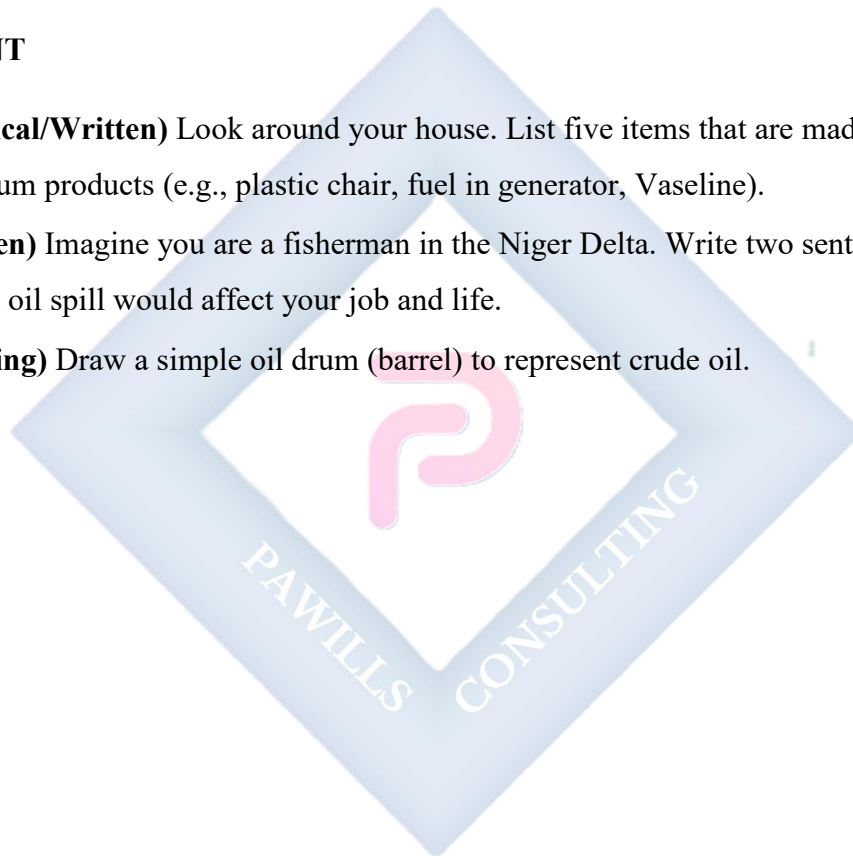
- **Non-Renewable Resources:** Minerals like crude oil, coal, and gold. They take millions of years to form. Once we use them up, they are gone forever.
- **Renewable Resources:** Resources that can be replaced naturally in a short time or are endless. *Examples:* Solar energy (sun), wind energy, trees (if replanted), water.
- **Sustainable Use:** We must use non-renewable minerals wisely and carefully. We should also invest in and use more renewable energy sources like solar power.

EVALUATION

1. How do mineral resources help the Nigerian government get money?
2. Name three useful products we get from refining crude oil.
3. What is an "oil spill" and why is it bad?
4. Is crude oil a renewable or non-renewable resource?
5. Why should we not waste mineral resources?
6. Give one example of a renewable energy source.

ASSIGNMENT

1. **(Practical/Written)** Look around your house. List five items that are made from or use petroleum products (e.g., plastic chair, fuel in generator, Vaseline).
2. **(Written)** Imagine you are a fisherman in the Niger Delta. Write two sentences about how an oil spill would affect your job and life.
3. **(Drawing)** Draw a simple oil drum (barrel) to represent crude oil.



PRIMARY 6 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Simple Machines: Levers I	Meaning of simple machines; Why we use simple machines - make work easier; Types of simple machines overview; Levers - definition and parts (fulcrum, effort, load); Types of levers - first class, second class, third class
2	Simple Machines: Levers II	Examples of first class levers - seesaw, scissors, pliers, crowbar; Examples of second class levers - wheelbarrow, nutcracker, bottle opener; Examples of third class levers - tweezers, fishing rod, broom; Mechanical advantage; Levers in the human body; Practical experiments
3	Simple Machines: Pulleys	Meaning of pulley; Parts of a pulley - wheel, rope/cable, axle; Fixed pulleys vs movable pulleys; Compound pulley systems; Uses of pulleys - flagpoles, cranes, elevators, wells; Mechanical advantage of pulleys; Making a simple pulley; Practical applications
4	Simple Machines: Inclined Plane	Meaning of inclined plane; How inclined planes work; Examples - ramps, slides, roads on hills, stairs, wedges, screws; Mechanical advantage of inclined planes; Effort vs load; Gradient and slope; Practical experiments; Applications in daily life and construction
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Primary and Secondary Colours I	Understanding colour; The colour spectrum; Primary colours - red, yellow, blue; Why they are called primary; Uses of primary colours; Secondary colours - orange, green, purple/violet; How secondary colours are formed
8	Primary and	Colour mixing experiments; Colour wheel; Complementary

	Secondary Colours II	colours; Warm and cool colours; Tertiary colours introduction; Colours in nature; Colours in art and design; Psychological effects of colours
9	Use of Drawing Instruments (Board Practice)	Types of drawing instruments - T-square, set squares (45° and 60°), compass, divider, protractor, drawing board, drawing pins/clips; Uses of each instrument; Care and maintenance; Drawing parallel lines, perpendicular lines; Drawing angles; Constructing shapes - triangles, rectangles, circles
10	Introduction to Woodwork Hand Tools	Types of woodwork hand tools - saw (hand saw, tenon saw), hammer, chisel, plane, file, screwdriver, measuring tape, try square, marking gauge; Uses of each tool; Parts of tools; Safety precautions in woodwork; Proper handling and storage; Basic woodwork techniques
11	Maintenance and Safety on Our Roads	Road signs and their meanings - warning signs, regulatory signs, information signs; Traffic lights; Zebra crossing; Road markings; Road safety rules for pedestrians; Road safety for cyclists; Safety in vehicles - seatbelts, child safety; Causes of road accidents; Preventing accidents; Road maintenance importance; Responsibilities of road users
12	EXAMINATION	End of term and final primary school assessment - theory and practical
13	CLOSING	Graduation ceremony; Report cards; Secondary school preparation; Celebration of achievements

PRIMARY 6 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

THIRD TERM

WEEK 1: SIMPLE MACHINES: LEVERS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a simple machine.
2. State the reason why we use simple machines.
3. Define a lever and identify its three main parts.
4. List the three classes of levers.

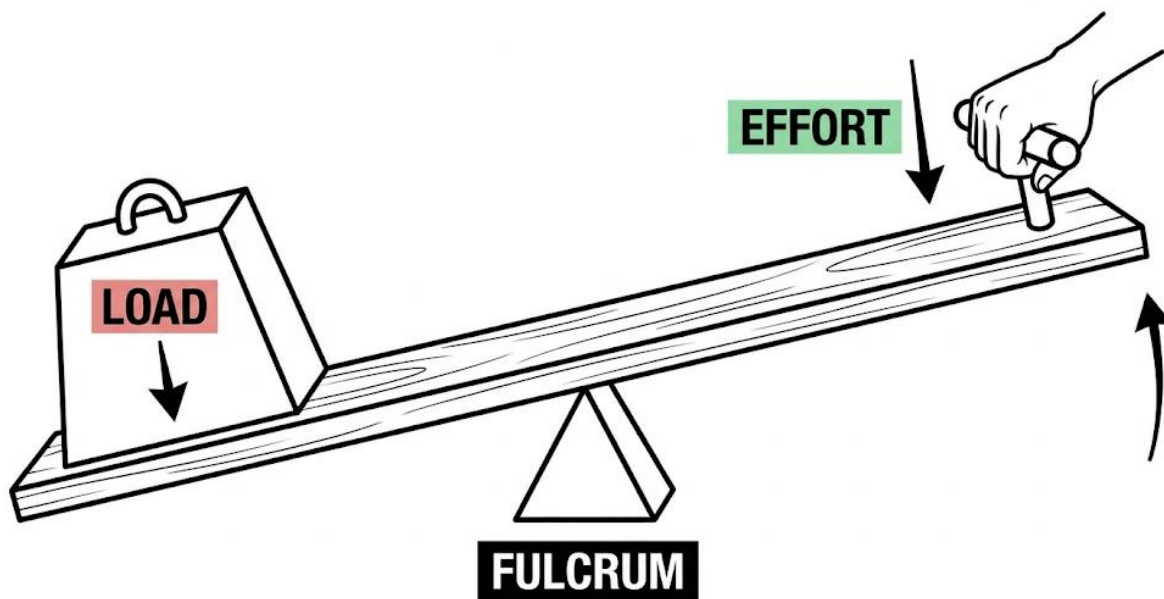
CONTENT

1. Meaning of Simple Machines A simple machine is a basic tool or device that makes our work easier. It helps us to:

- Use less effort (force) to move a heavy load.
- Change the direction of a force.
- Do work faster.
- *Example:* Using a bottle opener is easier than trying to open a bottle with your bare hands.

2. Levers A lever is one of the most common simple machines. It is a rigid bar (like a plank, rod, or stick) that turns or pivots on a fixed point. Every lever has three important parts:

1. **Fulcrum (Pivot Point) 'F':** The fixed point on which the lever turns or rests. It's like the support.
2. **Load 'L':** The weight or object we are trying to move or lift.
3. **Effort 'E':** The force we apply to move the load.



3. Classes of Levers Levers are grouped into three classes based on the position of the Fulcrum (F), Load (L), and Effort (E).

Class	Position of Middle Part	Example	Memory Trick
First Class	Fulcrum is in the middle.	See-saw, Scissors, Plier	F in middle = 1st
Second Class	Load is in the middle.	Wheelbarrow, Nutcracker, Bottle opener	L in middle = 2nd
Third Class	Effort is in the middle.	Broom, Tweezers, Fishing rod	E in middle = 3rd

Memory Aid: F - L - E (1 - 2 - 3). Whatever is in the middle gives the class number.

EVALUATION

1. What is the main purpose of a simple machine?
2. Name the three parts of a lever.
3. In a wheelbarrow, which part (F, L, or E) is in the middle?
4. What class of lever has the fulcrum in the middle?

5. Using the memory trick F-L-E, what does 'L' stand for?

ASSIGNMENT

1. **(Drawing)** Draw a simple lever (a stick on a triangle). Label the Fulcrum, Load, and Effort.
2. **(Written)** In your own words, explain what "effort" means when using a lever.
3. **(Written)** List the three classes of levers and give one example of each.



WEEK 2: SIMPLE MACHINES: LEVERS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify examples of First Class levers in the home/school.
2. Identify examples of Second Class levers.
3. Identify examples of Third Class levers.
4. Explain how levers are found in the human body.

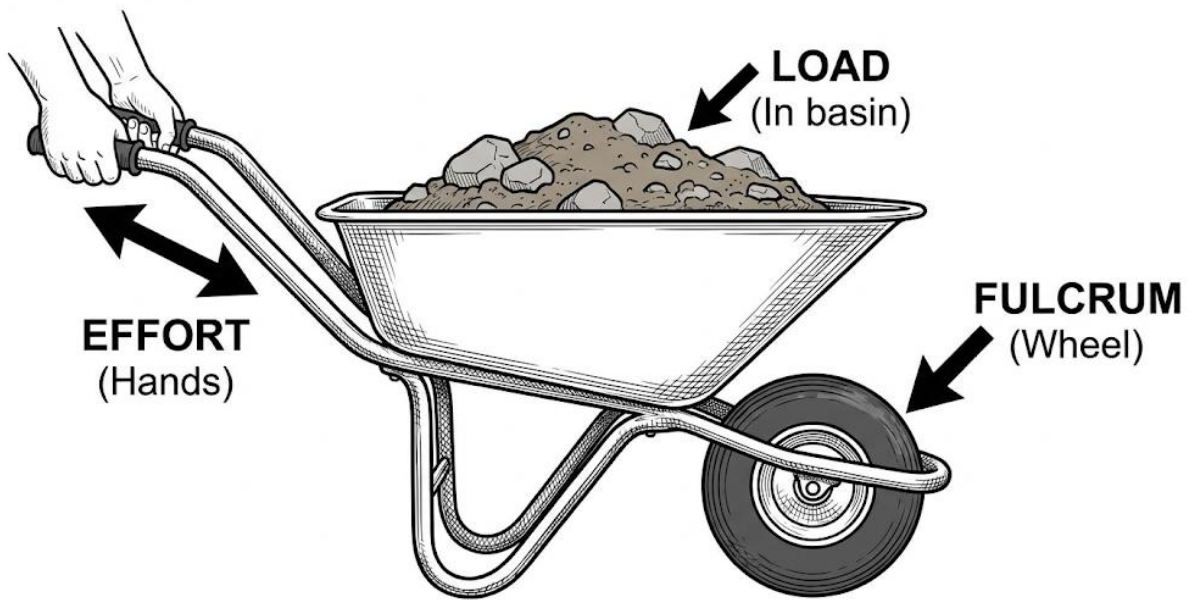
CONTENT

1. First Class Levers (Fulcrum in the Middle)

- **See-saw:** The support in the middle is the fulcrum. One child is the load, the other child pushing down is the effort.
- **Scissors/Plier:** The screw in the middle is the fulcrum. Your fingers apply effort, and the blades apply force on the paper/object (load).
- **Crowbar:** Used to lift heavy objects. The tip under the object is the fulcrum, you push down on the other end (effort) to lift the load.

2. Second Class Levers (Load in the Middle)

- **Wheelbarrow:** The wheel at the front is the fulcrum. The load (sand, cement) is in the middle bucket. You lift the handles at the back (effort).
- **Nutcracker:** The nut (load) is in the middle. The hinge is the fulcrum. You squeeze the handles (effort).
- **Bottle Opener:** The cap edge is the fulcrum, the cap is the load in the middle, you pull up the handle (effort).



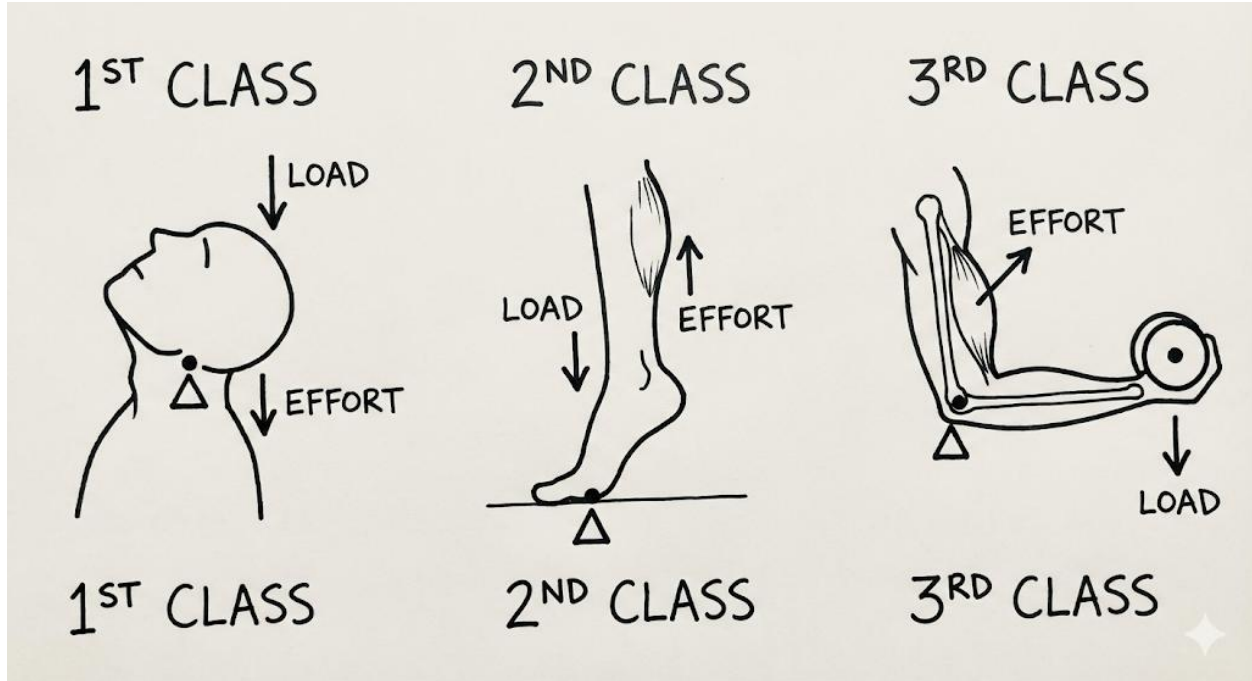
3. Third Class Levers (Effort in the Middle)

- **Broom:** Your top hand on the broomstick is the fulcrum. Your lower hand pushes in the middle (effort) to move the dust (load) at the bottom.
- **Tweezers/Tongs:** Your fingers squeeze in the middle (effort). The pivot at the top is the fulcrum. The tips pick up the object (load).
- **Fishing Rod:** Your hand at the rod's end is the fulcrum. You pull the rod in the middle (effort) to lift the fish (load) at the other end.
- **Your Arm:** When you lift a book, your elbow joint is the fulcrum. Your bicep muscle pulls in the middle of your forearm (effort) to lift the book in your hand (load).

4. Levers in the Human Body Our bodies are full of levers! Our bones are the rigid bars, and our joints are the fulcrums. Muscles provide the effort.

- **Nodding your head:** The neck joint is the fulcrum (F). The weight of your head is the load (L). Your neck muscles at the back provide the effort (E) – this is a First Class lever.
- **Standing on tiptoes:** The ball of your foot is the fulcrum (F). Your body weight is the load (L) in the middle. Your calf muscle pulls at the back (E) – this is a Second Class lever.

- **Bending your arm:** Your elbow is the fulcrum (F). The weight in your hand is the load (L). Your bicep muscle attaches in the middle of your forearm and pulls (E) – this is a Third Class lever.



EVALUATION

1. Is a pair of scissors a 1st, 2nd, or 3rd class lever? Why?
2. In a broom, where is the effort applied?
3. Give one example of a second-class lever used in the kitchen.
4. When you use a bottle opener, which part is the load?
5. What part of your body acts as the fulcrum when you kick a ball?
6. What class of lever is your arm when you are lifting a bucket of water?

ASSIGNMENT

1. **(Drawing)** Draw a pair of scissors. Mark and label the Fulcrum, Load (where it cuts), and Effort (where your fingers go).
2. **(Practical/Observation)** Find one tool in your kitchen that is a lever (e.g., spoon, fork, can opener). Say what class you think it is and why.

3. **(Written)** Explain in one sentence: "A wheelbarrow makes it easier to carry heavy sand because..."



WEEK 3: SIMPLE MACHINES: PULLEYS

OBJECTIVES By the end of the lesson, pupils should be able to:

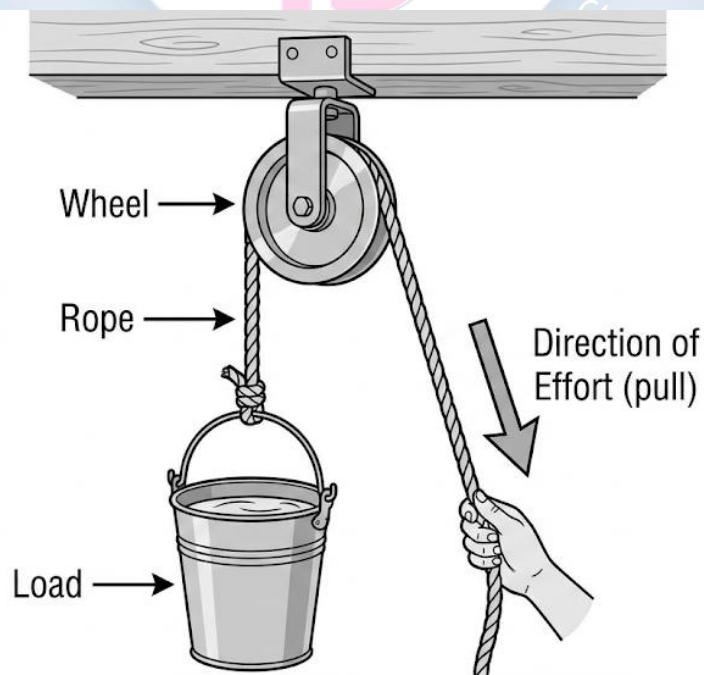
1. Define a pulley.
2. List the parts of a pulley system.
3. Differentiate between a fixed pulley and a movable pulley.
4. Mention three uses of pulleys in everyday life.

CONTENT

1. Meaning of a Pulley

A pulley is a simple machine made of a wheel with a groove (a cut) around its edge. A rope, chain, or cable runs through this groove.

- It is used to lift or lower heavy loads.
- How it works: You pull down on one end of the rope, and the load on the other end goes up.



2. Parts of a Pulley System

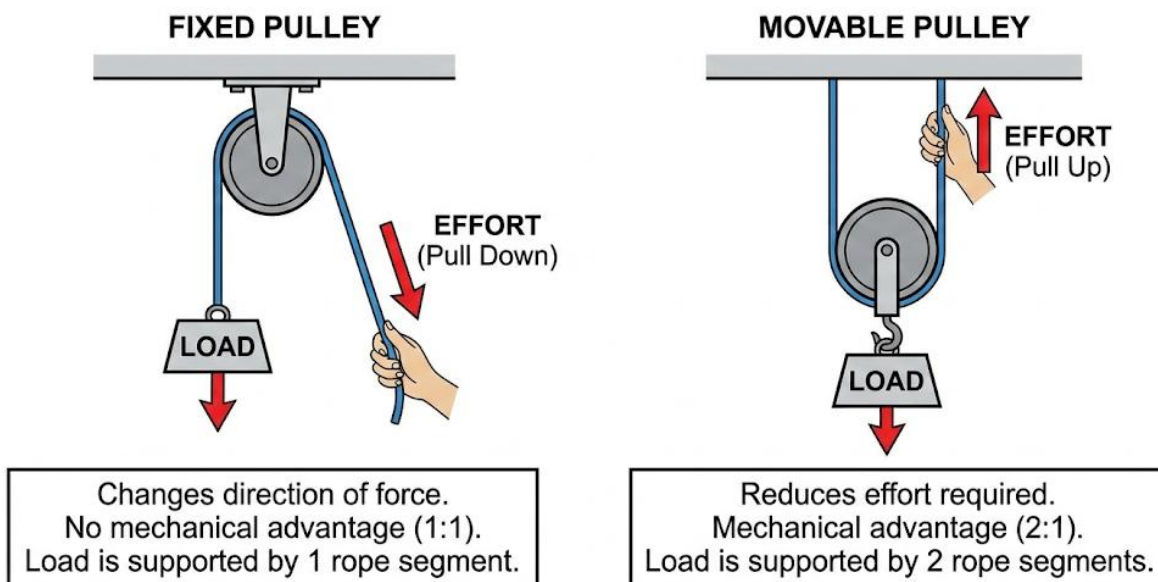
- **Wheel:** The round part that spins.

- **Axle:** The rod or pin in the center of the wheel that allows it to spin.
- **Rope/Cable:** The cord that goes around the wheel.
- **Load:** The object being lifted.
- **Effort:** The force applied to pull the rope.

3. Types of Pulleys

- **Fixed Pulley:**
 - The wheel is attached to a fixed support like a ceiling or beam.
 - *Advantage:* It changes the direction of the force. It is easier to pull down to lift something up than to pull it directly upwards.
 - *Disadvantage:* It does not make the load lighter. The effort needed is equal to the load's weight.
 - *Example:* A flagpole, a well with a fixed support, window blinds.
- **Movable Pulley:**
 - The wheel moves with the load. The rope is fixed at one end to a support.
 - *Advantage:* It makes the load feel lighter. The effort needed is about half the weight of the load.
 - *Disadvantage:* It does not change the direction of the force. You have to pull up to lift the load up.
 - *Example:* Some construction cranes, weight-lifting machines in gyms.
- **Compound/Block and Tackle Pulley:**
 - A combination of fixed and movable pulleys.
 - *Advantage:* It both changes direction and reduces effort significantly, making it very easy to lift extremely heavy loads.
 - *Example:* Cranes on construction sites, systems for lifting heavy engines in a garage.

Fixed vs. Movable Pulleys



4. Uses of Pulleys in Everyday Life

- Drawing water from a well.
- Raising and lowering flags on a flagpole.
- Lifting heavy goods in construction sites (cranes).
- Elevators (lifts) in tall buildings.
- Clotheslines (sometimes).
- Gym equipment.

EVALUATION

1. What is the main part of a pulley?
2. What is the advantage of using a fixed pulley?
3. Which type of pulley is attached to the load and moves with it?
4. Mention two places where you can find a pulley being used.
5. How does a pulley help a person drawing water from a deep well?

ASSIGNMENT

1. **(Drawing)** Draw a diagram showing a fixed pulley lifting a bucket of water from a well.
2. **(Written)** List three machines or places where you have seen pulleys being used.

3. **(Written)** Explain the main difference between a fixed pulley and a movable pulley.



WEEK 4: SIMPLE MACHINES: INCLINED PLANE

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define an inclined plane.
2. Explain how an inclined plane works.
3. Identify examples of inclined planes (ramps, stairs, screws).
4. State the relationship between the slope and the effort needed.

CONTENT

1. Meaning of Inclined Plane An inclined plane is a flat, slanted surface that connects a lower level to a higher level. It is also called a ramp.

- "Inclined" means slanted or tilted.
- "Plane" means a flat surface.

2. How it Works It is very hard to lift a heavy box straight up into a lorry. But if you use a strong, sloping plank (a ramp), you can push or roll the box up more easily.

- **Trade-off (The Simple Machine Rule):** The inclined plane lets you use less force (effort), but you have to move the object over a longer distance.
- It reduces effort by increasing distance.

Loading a Truck: Direct Lift vs Inclined Plane

1. Direct Lift



Lifting directly requires maximum effort against gravity. No mechanical advantage.

2. Inclined Plane



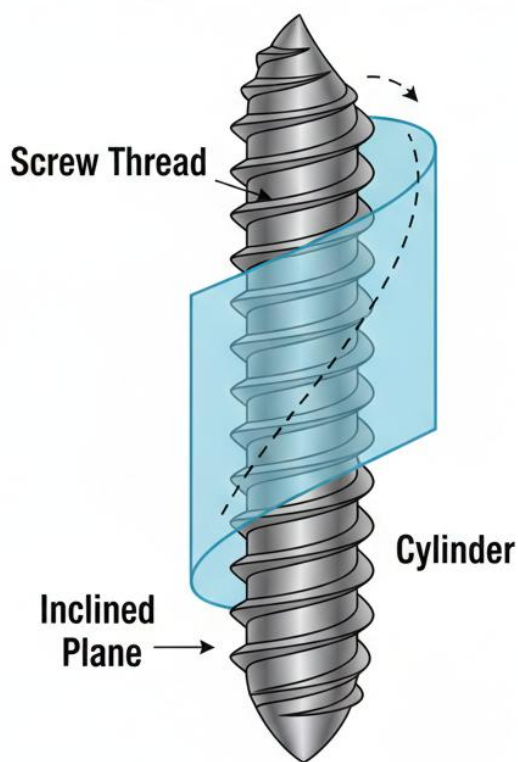
An inclined plane reduces effort needed by spreading the work over a longer distance. Simple machine!

3. Examples of Inclined Planes

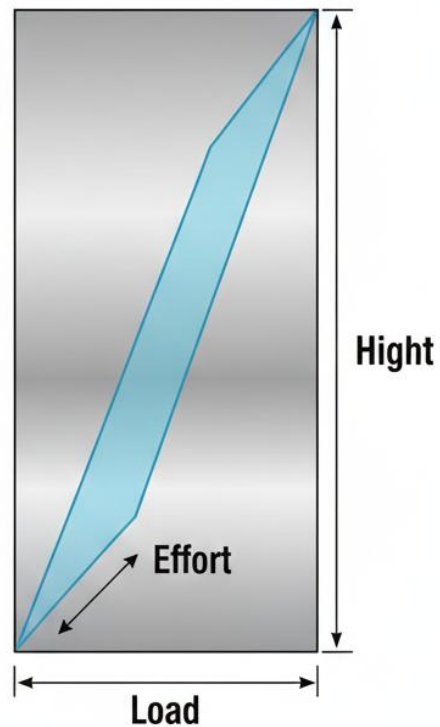
- **Ramps:** For wheelchairs to enter buildings, for loading goods into trucks.
- **Stairs and Ladders:** A staircase is a series of very short inclined planes.
- **Slides:** In the playground.
- **Wedges:** A wedge is two inclined planes back-to-back. *Examples:* Axe, knife, nail, doorstop. A wedge is used to split, cut, or hold things.

- **Screws:** A screw is an inclined plane wrapped around a cylinder (pole). The threads (spiral groove) are like a road winding up a hill. Turning the screw allows it to move into material with less effort than pushing a nail straight in.

THE SCREW: AN INCLINED PLANE



UNROLLED THREAD



4. Slope and Effort

- **Gentle Slope (Long Ramp):** Requires very little effort to climb, but you have to walk or push the object a long distance. *Example:* A long, gently sloping road up a hill.

- **Steep Slope (Short Ramp):** Requires more effort to climb, but the distance is short.

Example: A short, steep ladder.

EVALUATION

1. What is another name for an inclined plane?
2. Give two examples of an inclined plane found in your school.
3. Why is it easier to roll a drum up a ramp than to lift it?
4. Is a staircase an example of an inclined plane? Why?
5. Which is easier to push a heavy cart up: a long, gentle ramp or a short, steep ramp?
6. How is a screw related to an inclined plane?

ASSIGNMENT

1. **(Drawing)** Draw a picture of a person using a ramp to push a box into a truck.
2. **(Observation/Written)** Look closely at a screw or a bottle cap. Describe how the spiral groove looks like a winding road or a slide.
3. **(Practical/Written)** Look around your school or home. List two places where you see a ramp being used and say who uses it (e.g., ramp at the school entrance for everyone).

WEEK 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7: PRIMARY AND SECONDARY COLOURS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Primary Colours.
2. List the three primary colours.
3. Define Secondary Colours.
4. List the three secondary colours.

CONTENT

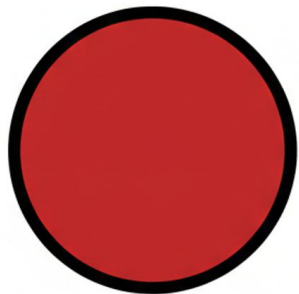
1. Understanding Colour

We see colour because of light. White light from the sun is actually a mixture of all colours (as seen in a rainbow). When light hits an object, some colours are absorbed and some are reflected. The colour that is reflected is the colour we see.

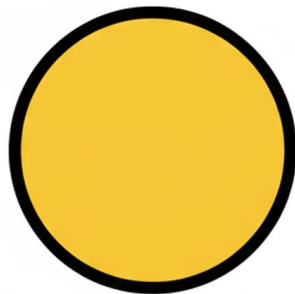
2. Primary Colours These are the basic, fundamental colours. They are called "primary" because:

- You cannot create them by mixing other colours together.
- All other colours are made from them.
- The three primary colours are: **RED, YELLOW, and BLUE.**

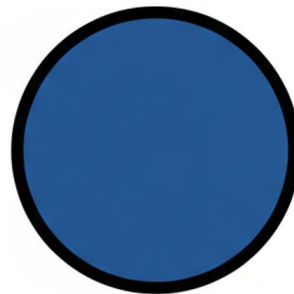
PRIMARY COLOURS



Red



Yellow



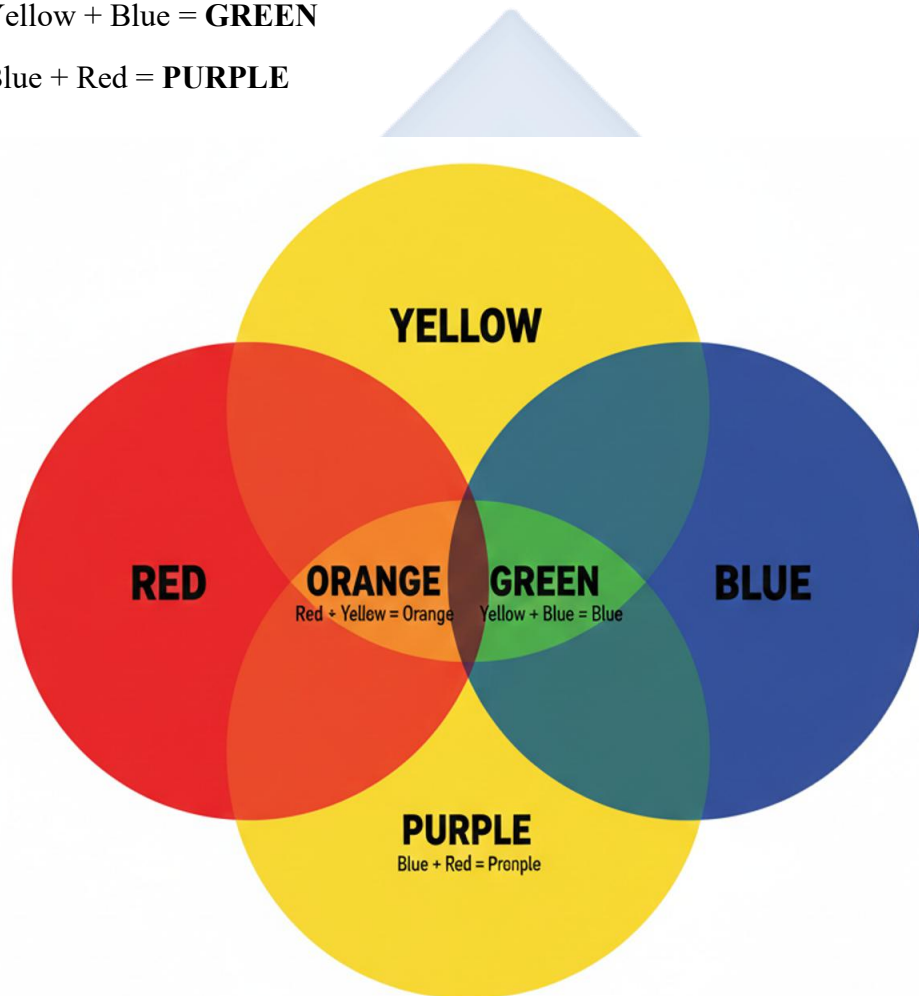
Blue

3. Secondary Colours These are colours you get by mixing two primary colours together in equal amounts.

- They are the "children" of the primary colours.
- The three secondary colours are: **ORANGE, GREEN, and PURPLE (VIOLET).**

4. Colour Mixing (The Colour Recipe)

- Red + Yellow = **ORANGE**
- Yellow + Blue = **GREEN**
- Blue + Red = **PURPLE**



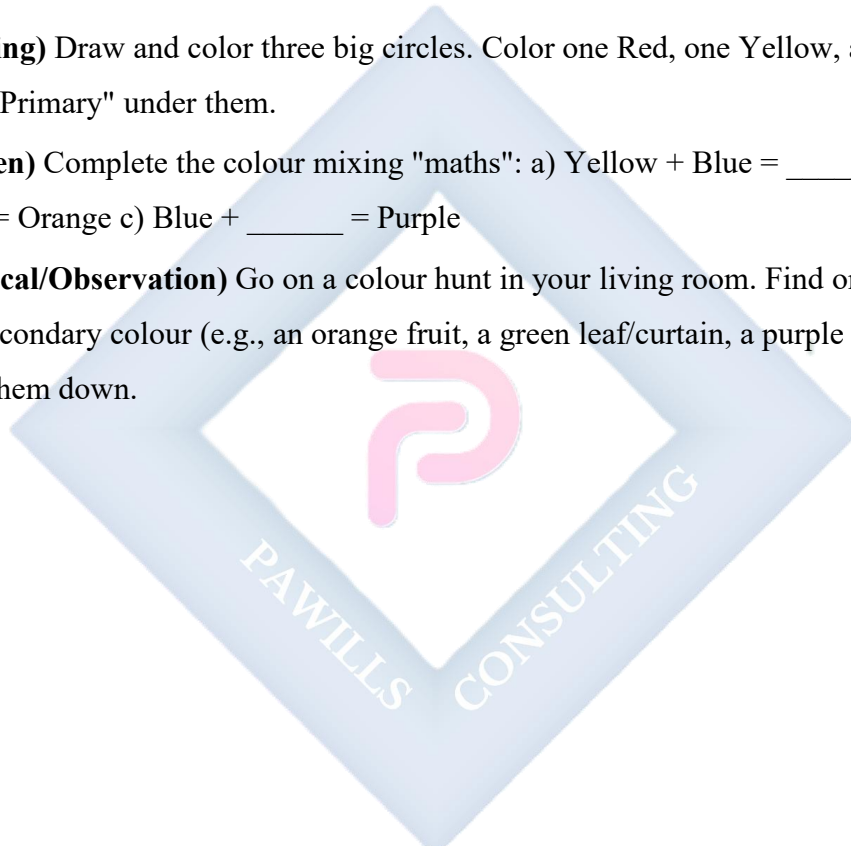
Important: Mixing light (like on a TV screen) uses different primary colours (Red, Green, Blue - RGB). We are talking about mixing paints, dyes, or crayons here.

EVALUATION

1. Why are Red, Yellow, and Blue called primary colours?
2. List the three primary colours.
3. Can you mix green and yellow to get red? (Yes/No)
4. What two colours do you mix to get green?
5. If you have red and blue paint, what colour will you make?
6. Name the three secondary colours.

ASSIGNMENT

1. **(Drawing)** Draw and color three big circles. Color one Red, one Yellow, and one Blue. Write "Primary" under them.
2. **(Written)** Complete the colour mixing "maths": a) Yellow + Blue = _____ b) _____ + Red = Orange c) Blue + _____ = Purple
3. **(Practical/Observation)** Go on a colour hunt in your living room. Find one object for each secondary colour (e.g., an orange fruit, a green leaf/curtain, a purple flower/book). Write them down.



WEEK 8: PRIMARY AND SECONDARY COLOURS II

OBJECTIVES By the end of the lesson, pupils should be able to:

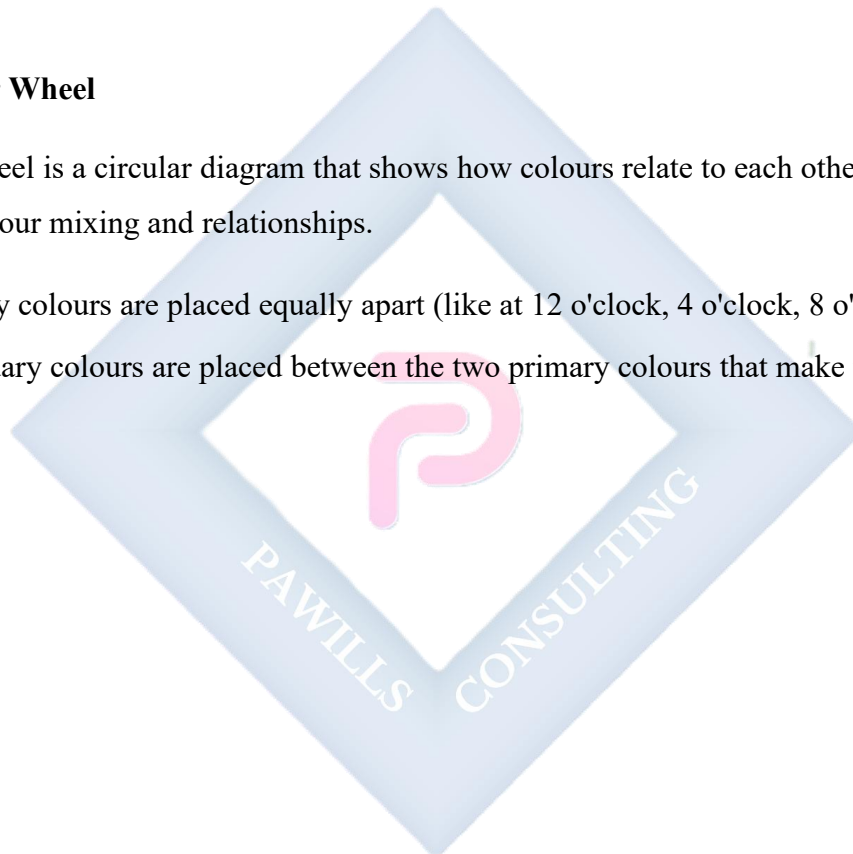
1. Create a colour wheel showing primary and secondary colours.
2. Identify warm and cool colours.
3. Explain what complementary colours are.
4. Mention one psychological effect of Red and Blue.

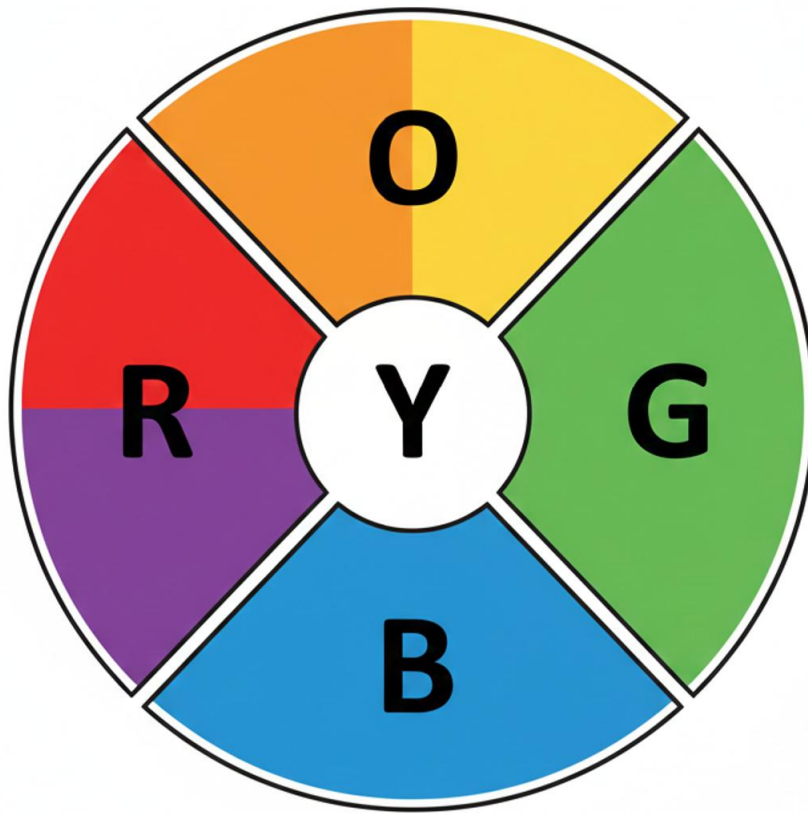
CONTENT

1. The Colour Wheel

The colour wheel is a circular diagram that shows how colours relate to each other. It helps us understand colour mixing and relationships.

- Primary colours are placed equally apart (like at 12 o'clock, 4 o'clock, 8 o'clock).
- Secondary colours are placed between the two primary colours that make them.





2. Warm and Cool Colours

Colours can affect how we feel. They are often grouped into:

- **Warm Colours:** Remind us of fire, sun, and heat. They feel energetic, bright, and exciting.
 - Red, Orange, Yellow.
 - *Example:* The sun, fire flame, ripe mango.
- **Cool Colours:** Remind us of water, sky, grass, and ice. They feel calm, peaceful, and relaxing.
 - Blue, Green, Purple.
 - *Example:* The ocean, the sky, a leafy tree.

3. Complementary Colours

These are pairs of colours that are opposite each other on the colour wheel. When placed side-by-side, they make each other look brighter and more vivid.

- Red is complementary to Green.
- Blue is complementary to Orange.
- Yellow is complementary to Purple.
- **Use:** Artists use them to make paintings "pop". They are also used in logos and designs to attract attention.

4. Psychology of Colour (How colours make us feel)

Colours can send messages and affect our moods:

- **Red:** Danger, Stop, Love, Energy, Excitement. (Fire engine, love heart).
- **Blue:** Peace, Calm, Trust, Sadness. (Sky, ocean, police uniform).
- **Yellow:** Happiness, Sunshine, Warning, Cheerfulness. (Sun, smiley face, caution sign).
- **Green:** Nature, Growth, Health, Safety. (Trees, grass, first aid sign).
- **Black:** Power, Mystery, Sadness.
- **White:** Purity, Cleanliness, Peace.

EVALUATION

1. On the colour wheel, which colour sits between Yellow and Blue?
2. Is Purple a warm or a cool colour?
3. Mention two warm colours.
4. What is the complementary colour of Blue?
5. Why might a hospital room be painted light blue or light green instead of bright red?
6. What feeling does the colour yellow often represent?

ASSIGNMENT

1. **(Drawing)** Draw and color a 6-part colour wheel correctly. Label all colours.
2. **(Reasoning/Written)** A restaurant wants to paint its walls. Would they choose bright red (energetic) or light blue (calm)? Give a reason for your choice.

3. **(Practical)** Look at the clothes you are wearing today. List the colours and say if they are mostly warm or cool colours.



WEEK 9: USE OF DRAWING INSTRUMENTS (BOARD PRACTICE)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify common drawing instruments (T-square, Set square, Compass).
2. State the use of each instrument.
3. Demonstrate how to draw parallel lines using a T-square and Set square.
4. Demonstrate how to draw a circle using a compass.

CONTENT

1. Drawing Instruments and Their Uses

For accurate technical drawings (like plans for buildings or machines), we use special tools.

Instrument	Picture/Description	Main Use
Drawing Board	A large, smooth, flat wooden board.	To hold the drawing paper firmly and provide a flat surface.
T-Square	A ruler shaped like a capital 'T'.	To draw straight horizontal lines. The head slides against the edge of the drawing board.
Set Squares	Clear plastic triangles (45° and 60°/30°).	To draw vertical lines and lines at specific angles (45°, 30°, 60°, 90°). Used with the T-square.
Compass	Two arms; one with a needle, one with a pencil lead.	To draw circles and arcs of specific sizes.
Divider	Looks like a compass but both ends have sharp points.	To transfer measurements from a ruler to a drawing or to divide lines into equal parts.
Protractor	A semi-circular tool with degrees marked from 0° to 180°.	To measure and draw angles.



PENCILS



PENCILS



CHARCOAL



INK PENS



INK PENS



PASTELS



MARKERS



MARKERS



CRAYONS



CRAYONS



BRUSHES



BRUSHES



RULER & TEMPLATES



COMPASS & PROTRACTOR

2. Care of Instruments

- Handle with care. Do not drop them.
- Keep the working edges of T-squares and set squares clean and free from nicks.
- Do not use the compass needle as a pin or toy; it is sharp.
- Store them neatly in a bag or box.

3. Basic Drawing Practices

- **Drawing Parallel Lines:**

- a. Place the T-square head firmly against the left edge of the board.
- b. Draw your first horizontal line along the T-square's blade.
- c. Slide the T-square up or down to a new position (keeping the head against the board).
- d. Draw the second line. They will be parallel.

- **Drawing Vertical/Angled Lines:**

- a. Place the T-square.
- b. Place a set square (e.g., 45° triangle) against the T-square's blade.
- c. Slide the set square along the T-square to the desired position.
- d. Draw your line along the vertical edge of the set square.

- **Drawing a Circle with a Compass:**

- a. Set the compass radius using a ruler (e.g., 4cm between needle tip and pencil tip).
- b. Place the needle point firmly where you want the circle's center.
- c. Hold the compass by the top knob and rotate it smoothly 360 degrees.
- d. A perfect circle is drawn.

EVALUATION

1. Which instrument is named after the letter 'T' and is used for horizontal lines?
2. What is the difference between a compass and a divider?
3. Which instrument would you use to draw a 60° angle?
4. What is the drawing board used for?
5. Why must we be careful with the sharp point of a compass?
6. How do you use a set square to draw a vertical line?

ASSIGNMENT

1. **(Practical Drawing)** Using a compass, draw a circle with a radius of 5cm.
2. **(Practical Drawing)** Use a ruler to draw a rectangle of length 7cm and width 4cm.
3. **(Written)** Write one sentence for each: a) What a T-square does. b) What a protractor does.

WEEK 10: INTRODUCTION TO WOODWORK HAND TOOLS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List five common woodwork hand tools.
2. Identify measuring, cutting, and driving tools.
3. State the safety precautions when using hand tools.
4. Describe the correct way to hold a saw.

CONTENT

1. Types of Woodwork Hand Tools Tools are grouped by their main function.

Group	Tool Name	Picture/Description	Main Use
Measuring & Marking	Tape Measure / Rule	A flexible ruler that rolls up.	Measuring long pieces of wood.
	Try Square	An 'L' shaped metal tool.	Checking if corners are square (90°) and marking straight lines across wood.
Cutting	Hand Saw	A long saw with a wide blade.	Cutting wood along or across the grain.
	Tenon Saw	A saw with a stiff back (a strip of metal).	Cutting straight lines and joints accurately.
	Chisel	A metal blade with a sharp edge and a handle.	Carving, shaping, or cutting small pieces of wood.
	Plane	A tool with a sharp blade set in a wooden/metal body.	Smoothing and flattening rough wood surfaces.
Driving & Striking	Claw Hammer	A hammer with a flat head and a forked claw.	Driving nails in and pulling nails out.
	Mallet	A wooden hammer.	Hitting chisels (softer than a metal hammer).

	Screwdriver	A tool with a shaped tip (flat or star).	Turning screws into wood.
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COMMON WOODWORK TOOLS



Hammer



Saw



Screwdriver



Chisel



Try Square

2. Safety Precautions (VERY IMPORTANT)

- Always carry sharp tools with the sharp edge pointing downwards and away from your body.
- Never put tools in your pocket.
- Keep your hands behind the cutting edge of saws and chisels.

- Wear strong shoes (not sandals) in the workshop in case a tool falls.
- Use tools only for their intended purpose.
- Store tools properly in a toolbox when not in use. Do not leave them lying around.
- Always cut away from your body, not towards it.

3. Correct Way to Hold and Use a Hand Saw

- **Grip:** Hold the handle firmly with your dominant hand.
- **Guide:** Place your other hand on the wood to guide the saw, but keep fingers clear of the blade.
- **Starting the Cut:** Use your thumb to guide the blade at the start. Make a few gentle backward strokes to create a small groove.
- **Sawing:** Use long, steady, forward strokes. Let the saw do the work; don't push too hard. Keep the saw straight.

EVALUATION

1. Name one tool used for measuring wood.
2. What is the main difference between a hand saw and a tenon saw?
3. Which tool would you use to make a wooden surface smooth?
4. What is a try square used for?
5. Mention two important safety rules when using hand tools.
6. Why is a mallet used with a chisel instead of a metal hammer?

ASSIGNMENT

1. **(Drawing)** Draw a hammer and label its two main parts: Head and Handle.
2. **(Drawing)** Draw the shape of a hand saw.
3. **(Written)** Write one sentence explaining why you should never run in a place where people are using tools.

WEEK 11: MAINTENANCE AND SAFETY ON OUR ROADS

OBJECTIVES By the end of the lesson, pupils should be able to:

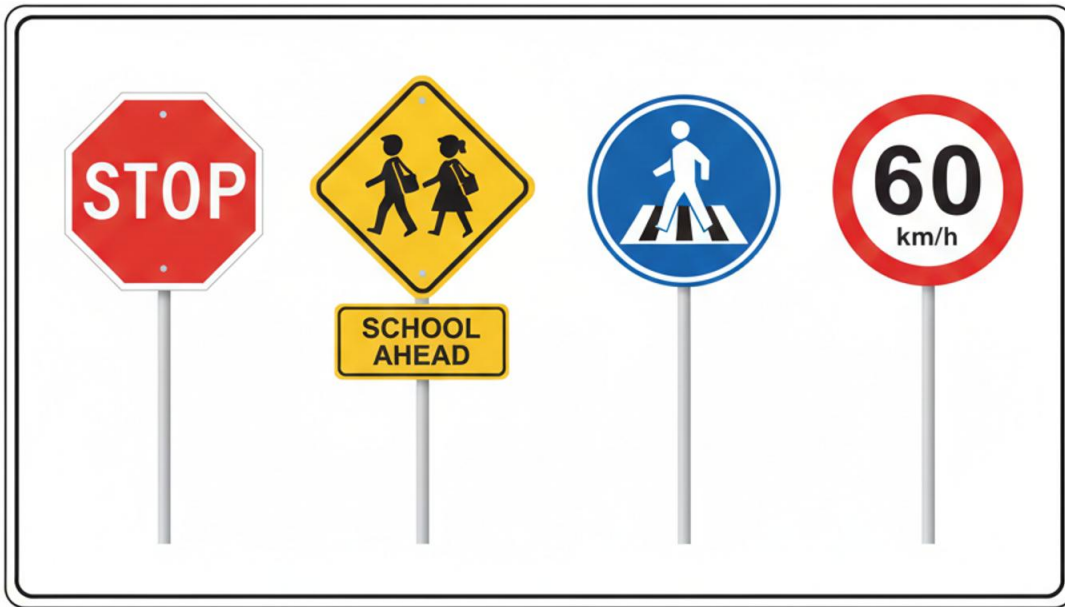
1. Interpret common road signs (Stop, Zebra Crossing, Speed Limit).
2. Explain the meaning of the three Traffic Light colours.
3. List three causes of road accidents.
4. Mention three safety rules for pedestrians and cyclists.

CONTENT

1. Road Signs and Their Meanings

Road signs give information, warnings, and orders to road users (drivers, cyclists, walkers).

Sign Type	Shape & Color	Example & Meaning
Regulatory (Mandatory)	Circular, Red/White or Blue/White	STOP (Octagon, Red): Must come to a complete halt. NO ENTRY: Do not enter that road. SPEED LIMIT 50: Do not drive faster than 50km/h.
Warning	Triangular, Red border/White	SCHOOL AHEAD: Warns drivers that children may be crossing. BEND AHEAD: Warns of a curve in the road. BUMP/ROAD HUMPS: Warns of bumps on the road.
Informational/Guide	Rectangular, Various colors	HOSPITAL: Direction to a hospital. PARKING: You can park here. ZEBRA CROSSING: Shows location of a pedestrian crossing.



2. Road Markings

- **Zebra Crossing:** Black and white stripes on the road. Pedestrians have the right of way here. Vehicles must stop if someone is waiting to cross or is crossing.
- **White/Yellow Lines:** Separate lanes of traffic. A solid line means you should not cross it.

3. Traffic Lights

- **RED means STOP.** Wait behind the stop line.
- **YELLOW/AMBER means GET READY.** It warns that the light is about to change to red. Stop if safe to do so.
- **GREEN means GO.** You may proceed with caution.

4. Causes of Road Accidents

- **Overspeeding:** Driving too fast for the road conditions.
- **Distracted Driving:** Using a phone, eating, or talking while driving.
- **Drunk Driving:** Driving under the influence of alcohol or drugs.
- **Bad Roads:** Potholes, eroded edges, lack of signs.

- Poor Vehicle Maintenance: Bad brakes, worn-out tyres, faulty lights.
- Disobeying Traffic Rules: Jumping red lights, ignoring signs.

5. Safety Rules for Different Road Users

- **For Pedestrians (Walkers):**

- Always walk on the pavement (sidewalk) if there is one.
- If no pavement, walk facing oncoming traffic (on the right side of the road).
- Cross at safe places: Use Zebra crossings, footbridges, or pedestrian traffic lights.
- Look Right, Left, and Right again before crossing. Keep looking as you cross.
- Do not run across the road.

- **For Cyclists:**

- Wear a helmet to protect your head.
- Use hand signals to show when you are turning left or right.
- Ride in single file, not side-by-side.
- Ensure your bicycle has working brakes, lights, and reflectors.

- **For Motorists (Drivers):**

- Wear seatbelts always.
- Do not speed.
- Regular maintenance: Check tyre pressure, brakes, lights, and engine oil regularly.
- Never drive when tired or sleepy.

EVALUATION

1. What must you do when you see a RED traffic light?
2. What shape are most warning signs?
3. What does a "Zebra Crossing" look like and who should use it?
4. Mention two things a driver might do that could cause an accident.
5. Why is it important for a cyclist to wear a helmet?
6. What should you do before stepping onto a road to cross it?

ASSIGNMENT

1. **(Drawing)** Draw and color a set of traffic lights (Red, Yellow, Green). Write what each color means underneath.
2. **(Drawing)** Draw the road sign for "STOP" or "SCHOOL AHEAD".
3. **(Written)** List three things a good driver should regularly check on their car to keep it safe (e.g., Brakes, Tyres, _____).

